

AA SED MSc + MArch Sustainable Environmental Design 2024-25

Architectural Association School of Architecture

Term 1 Project

Refurbishing the City: London Building Studies

Atmos Lab Office

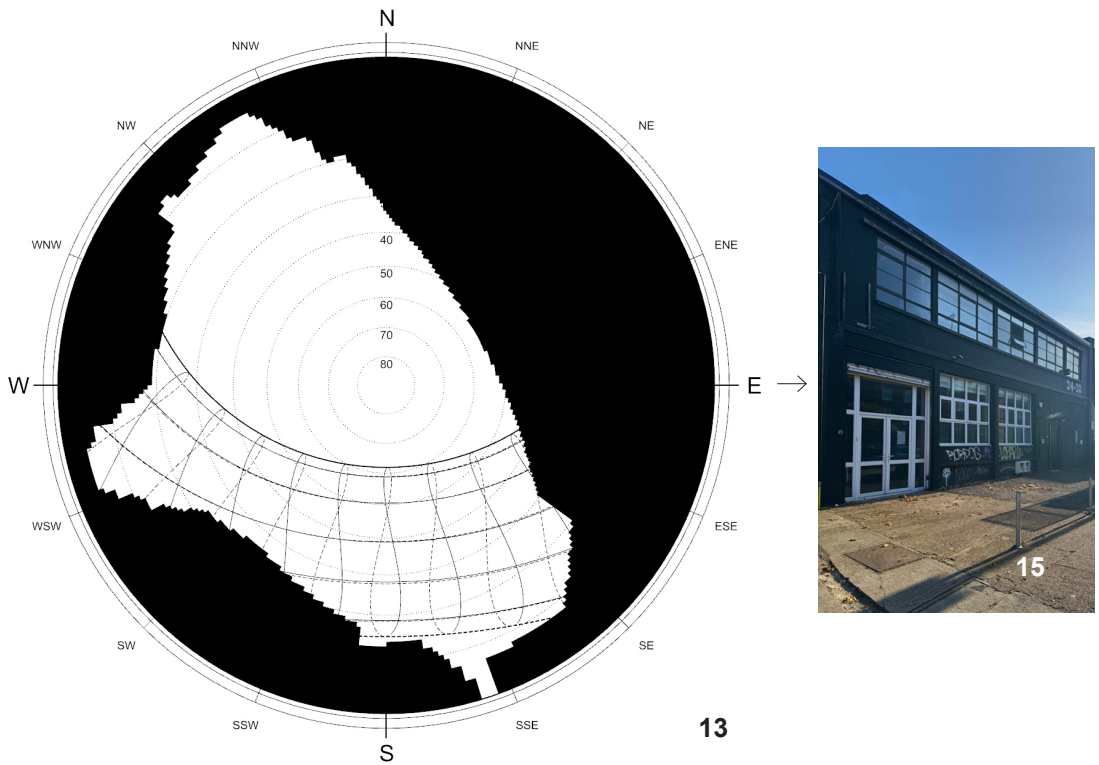
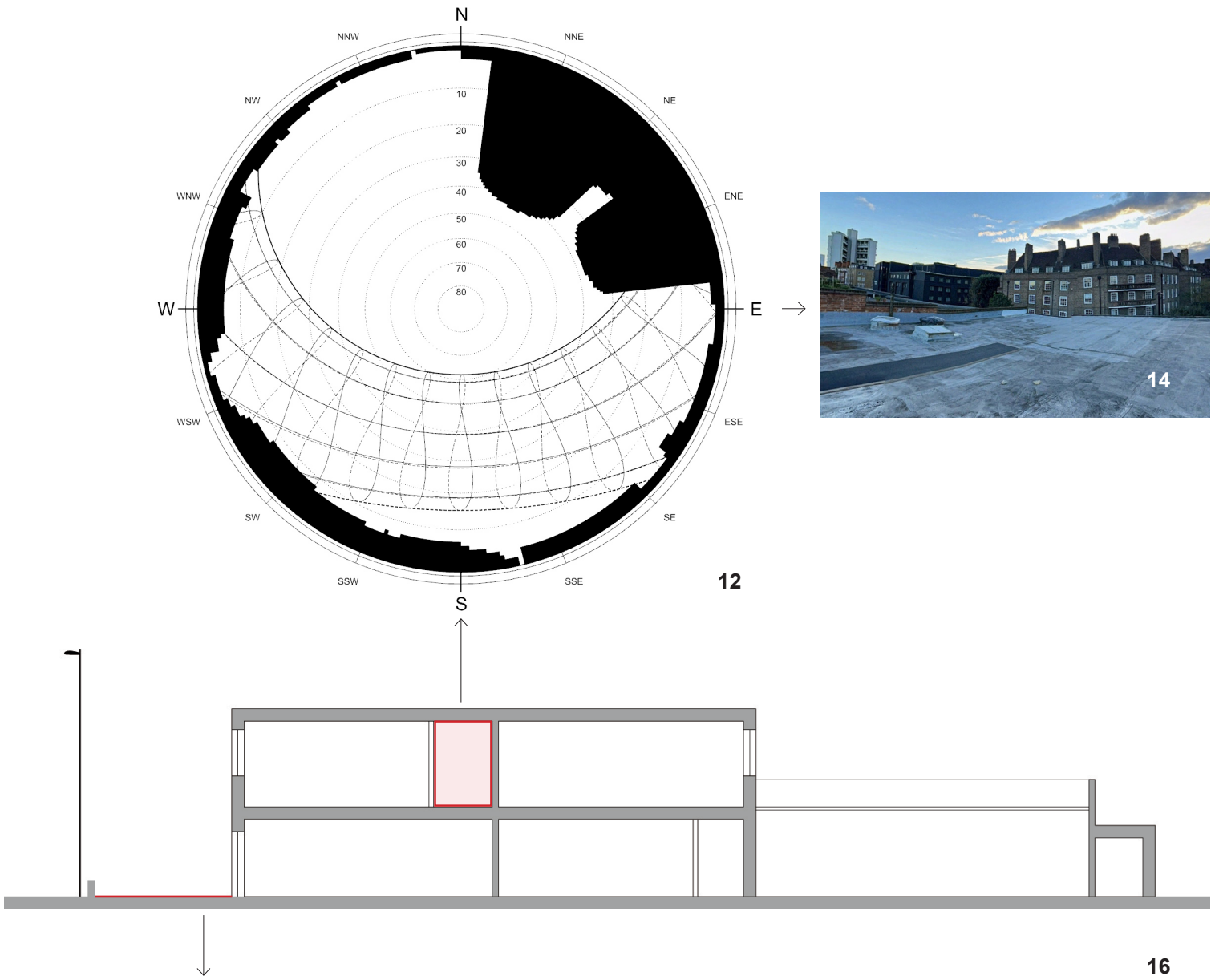
Mia Morrone

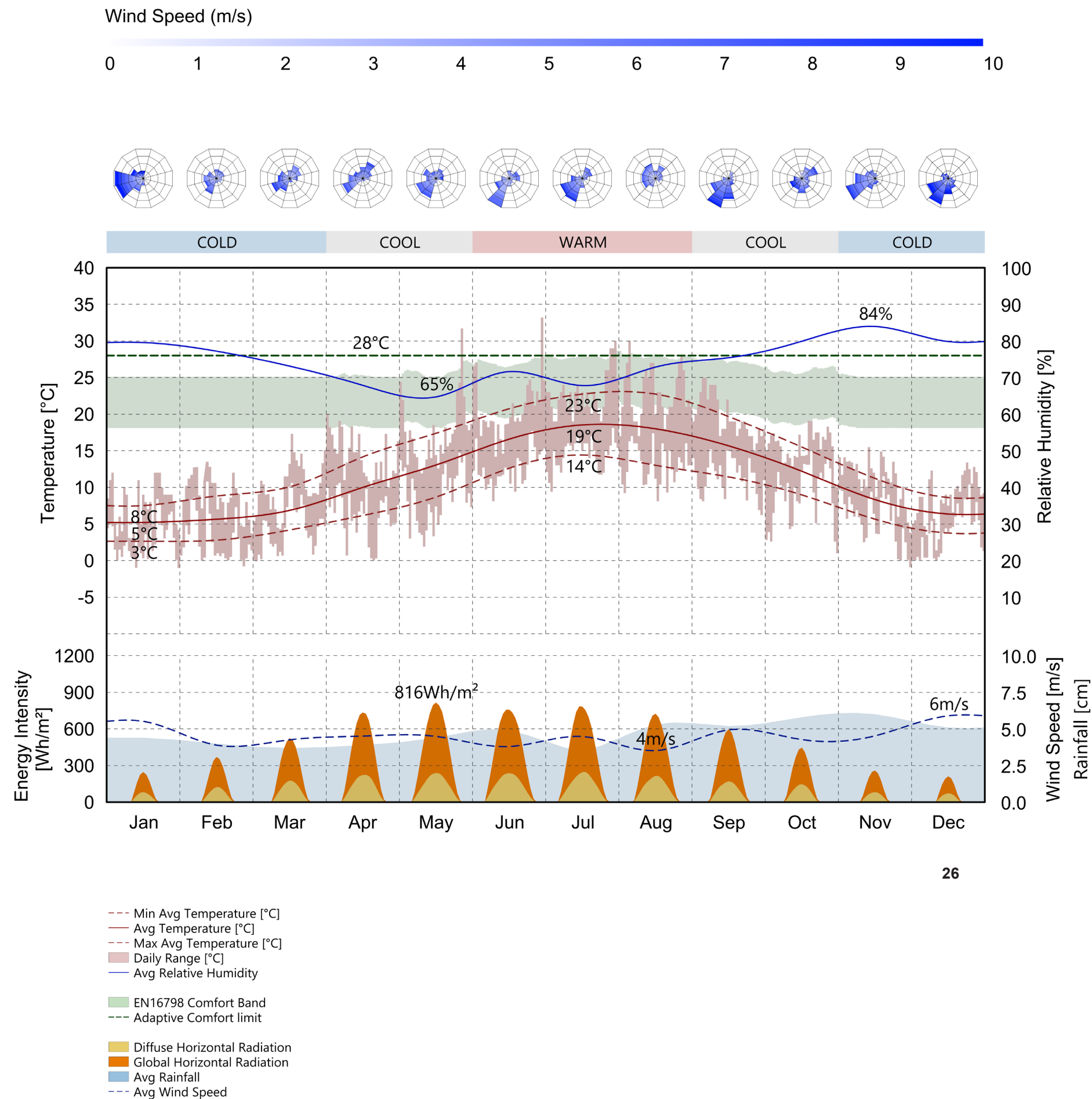
January 2025

2.3. Case

ATMOS LAB

The Atmos Lab office overlooks the Broadway East project by the Berkeley Group, a development of towers centered around the Bethnal Green gasholders. As shown in Fig. 12, these new constructions cast minimal shadows on the building's roof. Both the tenants of Atmos Lab and their neighbors have shown interest in using the entrance and roof areas. They recognize the potential of these spaces to improve user's comfort. The roof remains unshaded throughout the day, while the entrance is shaded during early morning and late afternoon hours due to nearby structures. However, it benefits from direct sunlight around midday year-round.





2.4. Climate

CLIMATE ANALYSIS

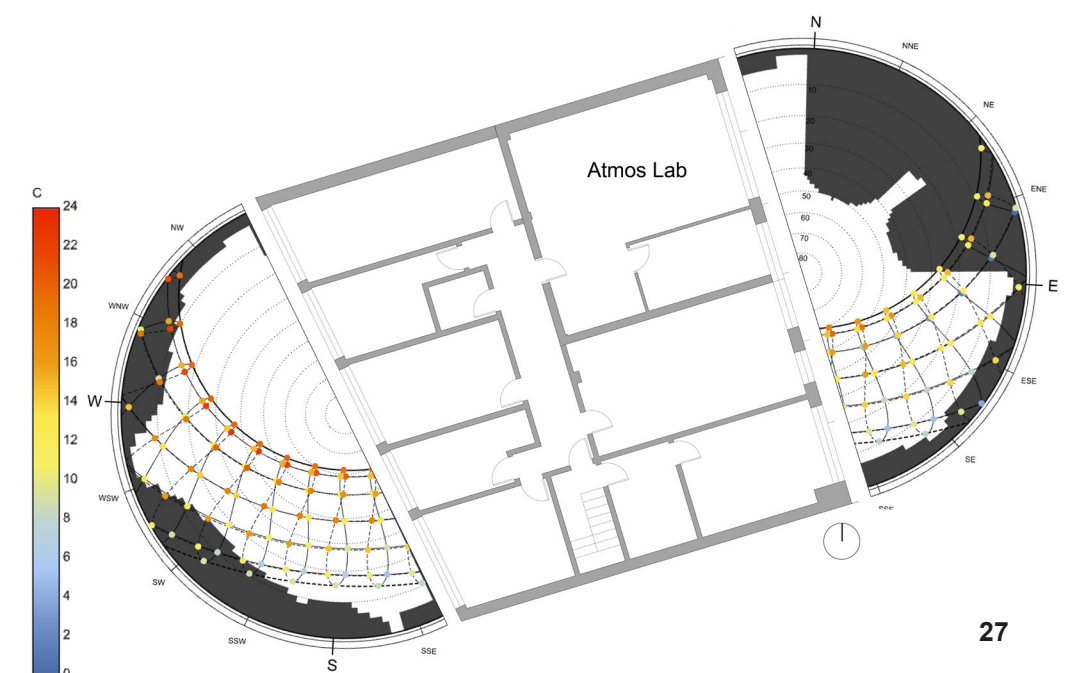
According to the Köppen-Geiger climate classification and the yearly climate chart, London has a temperate maritime climate characterized by mild winters and moderately warm summers, with no dry season.

- Mean Annual Temperature: 11°C
- Temperature Range: -1°C to 33°C
- Days Above 28°C: 31 (less than 1% of the year)
- Average Daily Temperature Difference (ΔT): 7°C
- Precipitation: Frequent but low intensity, averaging 36-60 mm monthly
- Relative Humidity: 65-84%
- Prevailing Wind: Southwest, with speeds ranging from 4 m/s in warmer months to 6 m/s in colder months
- Solar Radiation: 600-800 Wh/m² from April to August; below 300 Wh/m² from November to January

Three distinct seasons were identified: warm, transitional, and cold.

SUN PATH & SHADING MASK

Both façades receive sunlight for at least half of the day throughout all seasons. The southwest side (front façade) is most exposed to the sun during midday and afternoon, while the northeast side (rear façade) primarily receives direct sunlight during the early morning hours and at noon.



3.2. Spot Measurements

DAYLIGHT

The illuminance levels were measured in two sections of Atmos Lab: one across the main office area and the other across the meeting room and kitchen area. The measurements were taken at midday under a partly cloudy sky. The results are plotted in Fig. 31 and 32, where the 300 lux threshold is highlighted. The measurements confirmed the occupants' complaints about the lack of daylight on the two tables furthest from the window and the sufficient daylight levels in the meeting room.

The office is equipped with silver solar curtains (Fig. 33) designed by the architectural practice Lacaton & Vassal. These curtains have a reflective Mylar side facing outward to reflect undesired solar radiation and a fabric side facing inward. Since this is a primary adaptive strategy used by the occupants to reduce solar gains and filter daylight during warm months, the illuminance level near the window was measured both with and without the curtain to estimate its transmittance. The obtained value was 56%.

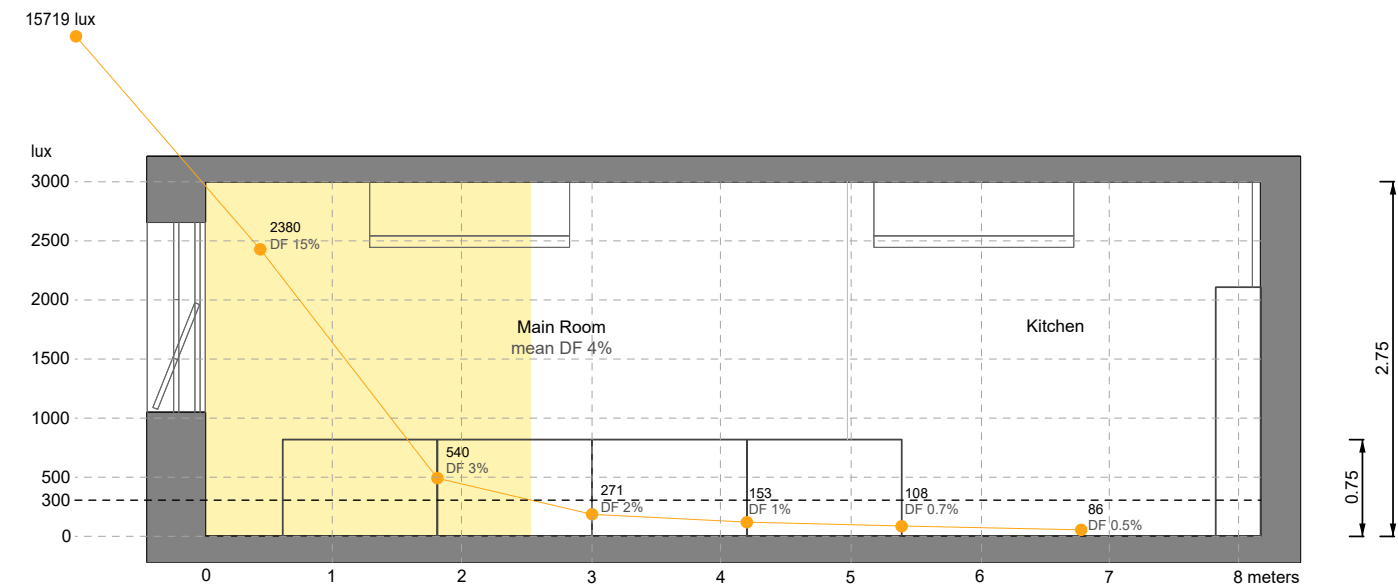


Fig. 31
Office section with measured illuminance levels at 0.75m from the floor.

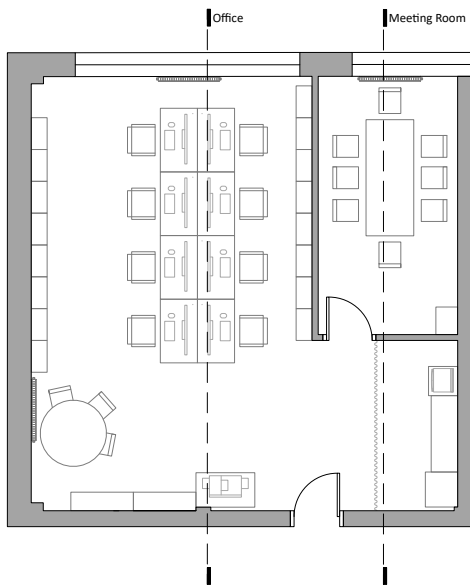
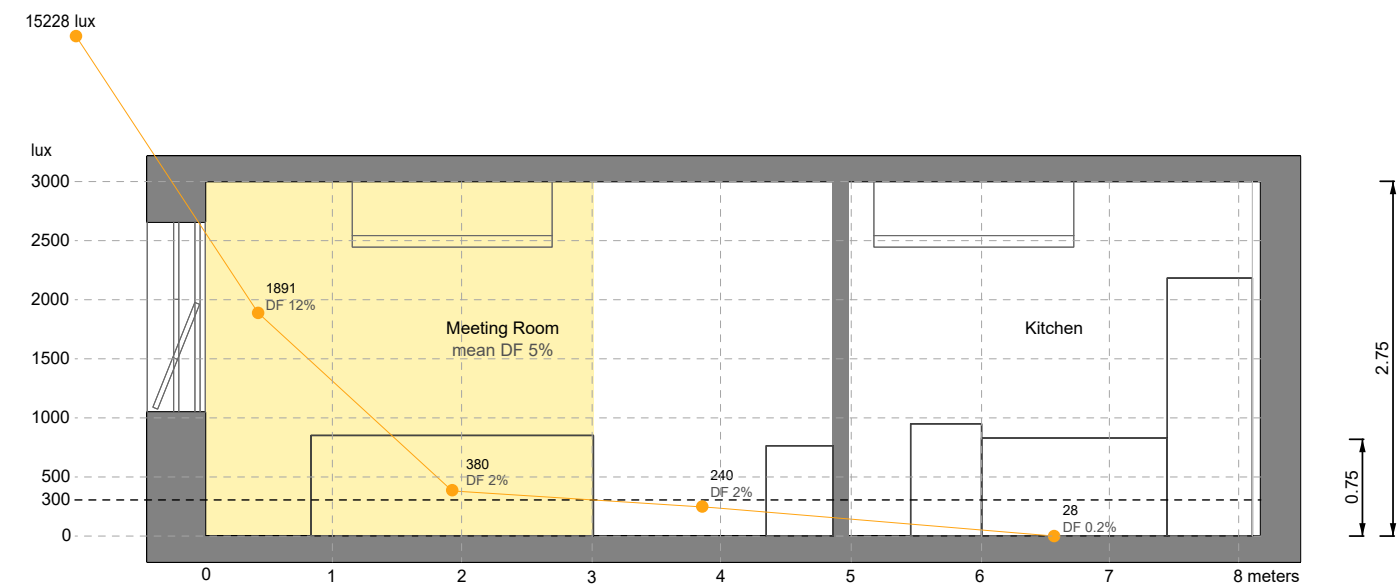


Fig. 30
Section location plan.

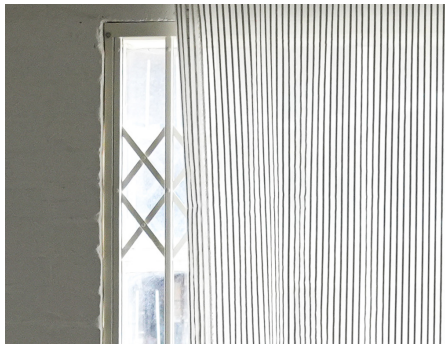


Fig. 33
Picture of the silver solar curtain installed in Atmos Lab Office.



Fig. 34
Picture of Atmos Lab main office with measured environmental parameters.

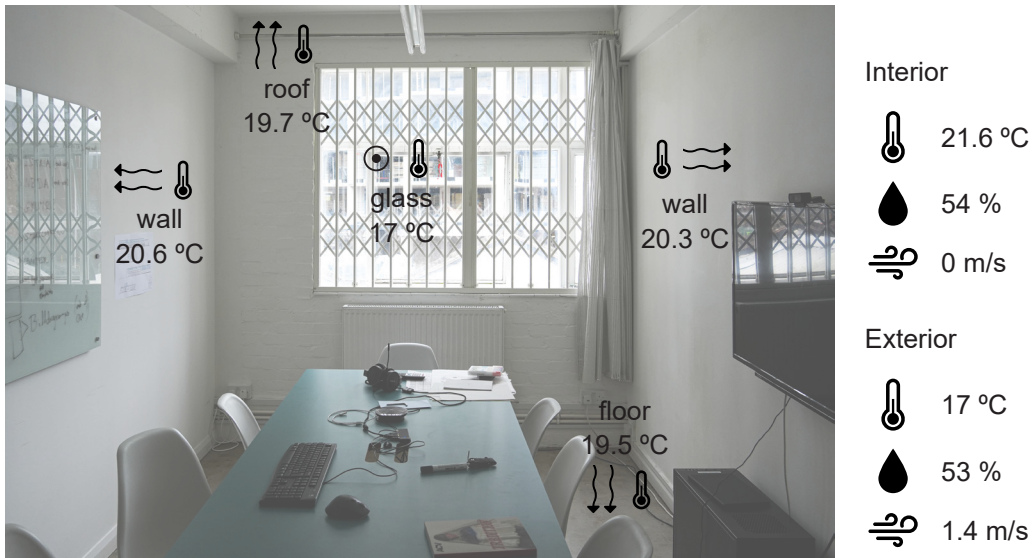
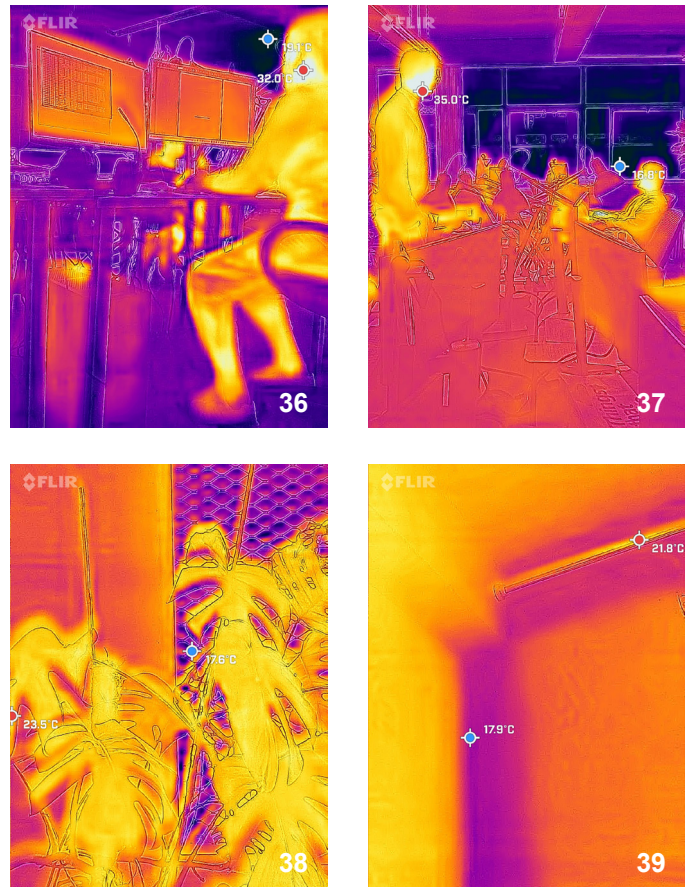


Fig. 35
Picture of Atmos Lab meeting room with measured environmental parameters.



OTHER ENVIRONMENTAL PARAMETERS

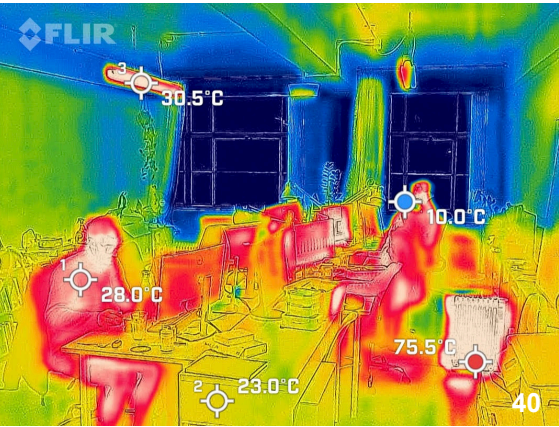
The spot measurements of internal and external temperature, relative humidity, wind speed, and the radiant temperature of relevant surfaces were indicated on pictures of the main office and meeting room in Atmos Lab (Figs. 34 and 35). Since the windows are single-glazed, the radiant temperature of the glass can be approximated by the external air temperature. The measured data aligns with the thermal images taken by the group using Atmos Lab's thermal camera (Figs. 36 to 39). Additionally, a thermal image with an alternative colour scale (Fig. 40), taken by Olivier Dambron during a winter day, clearly shows the radiant temperatures of the main office's surfaces, equipment, occupants, and portable heater.

The measurements corroborate the group's observations during the site visit: the low radiant temperature of the glass and air infiltration through window gaps make the two office spaces near the window colder. Additionally, the contrasting temperature felt on either side of the face when seated can cause thermal discomfort. The external wall is two degrees colder than the insulated and unexposed surfaces. The computers exhibit a higher radiant temperature, which can either positively or negatively affect occupants, depending on the season. Finally, the thermal image in Fig. 39 reveals a thermal bridge caused by the concrete structure in the façade.

Fig. 36, 37, and 38
Thermal images showing the radiant temperature of equipment, people, and windows.

Fig. 39
Thermal image highlighting the thermal bridge caused by the concrete structure in the external wall.

Fig. 40
Thermal image of Atmos Lab main office (source: Olivier Dambron)



WEEK GRAPH

Three data loggers were positioned in distinct locations: the meeting room, the main office area, and the rooftop of a neighbouring building. These devices recorded air temperature and relative humidity at 30-minute intervals between 10 and 17 October 2024, with the results presented in Figs. 41 and 45. Additionally, solar radiation and rainfall data were obtained from a nearby weather station in Bow, London, via the London Grid for Learning Trust website. Furthermore, CO2 concentrations were measured using an Atmos Lab data logger, with data provided by Olivier Dambron. The placement of all data loggers is illustrated in Figs. 22 and 42 to 44.

Analysis of the weekly graph revealed three spikes caused by direct solar radiation striking the main office data logger between 07:30 and 08:30, which should be disregarded. Additionally, an increase in temperature was observed in the meeting room on two nights, likely due to a high-performance computer running simulations overnight.

It is also worth mentioning that internal temperatures remained within the comfort band throughout the analysed period, with noticeable increases during working hours, as evidenced by a comparison of weekdays with weekends. This suggests that internal temperature rises are primarily driven by internal heat gains. Furthermore, internal temperatures did not drop below 19°C, even during unoccupied periods. This is likely due to the building's central heating system, which is managed by the building management company. Although the radiators in Atmos Lab were turned off during the measurement period, heat transfer from adjacent offices, where radiators may have been in use, likely contributed to maintaining higher internal temperatures.

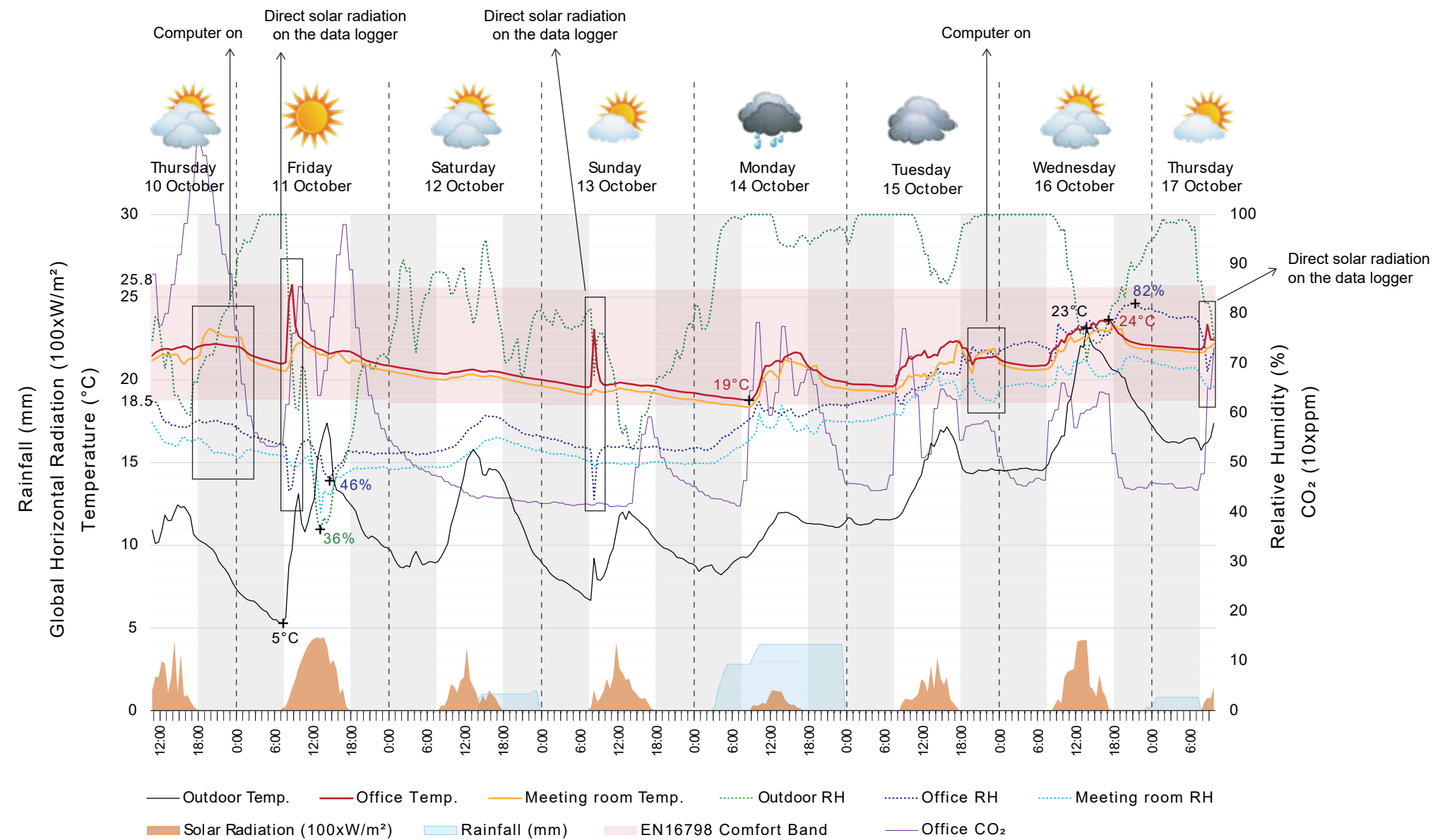
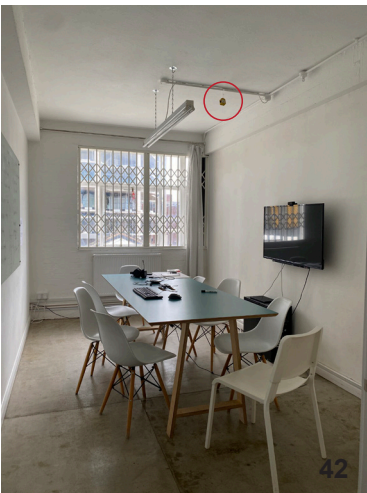


Fig. 41
Graph illustrating data obtained from data loggers and Bow's weather station.

Fig. 42
Location of the datalogger in the meeting room.

Fig. 43
Location of the datalogger in the main office.

Fig. 44
Location of the datalogger in the rooftop of the neighbour building.



TUESDAY 15 October

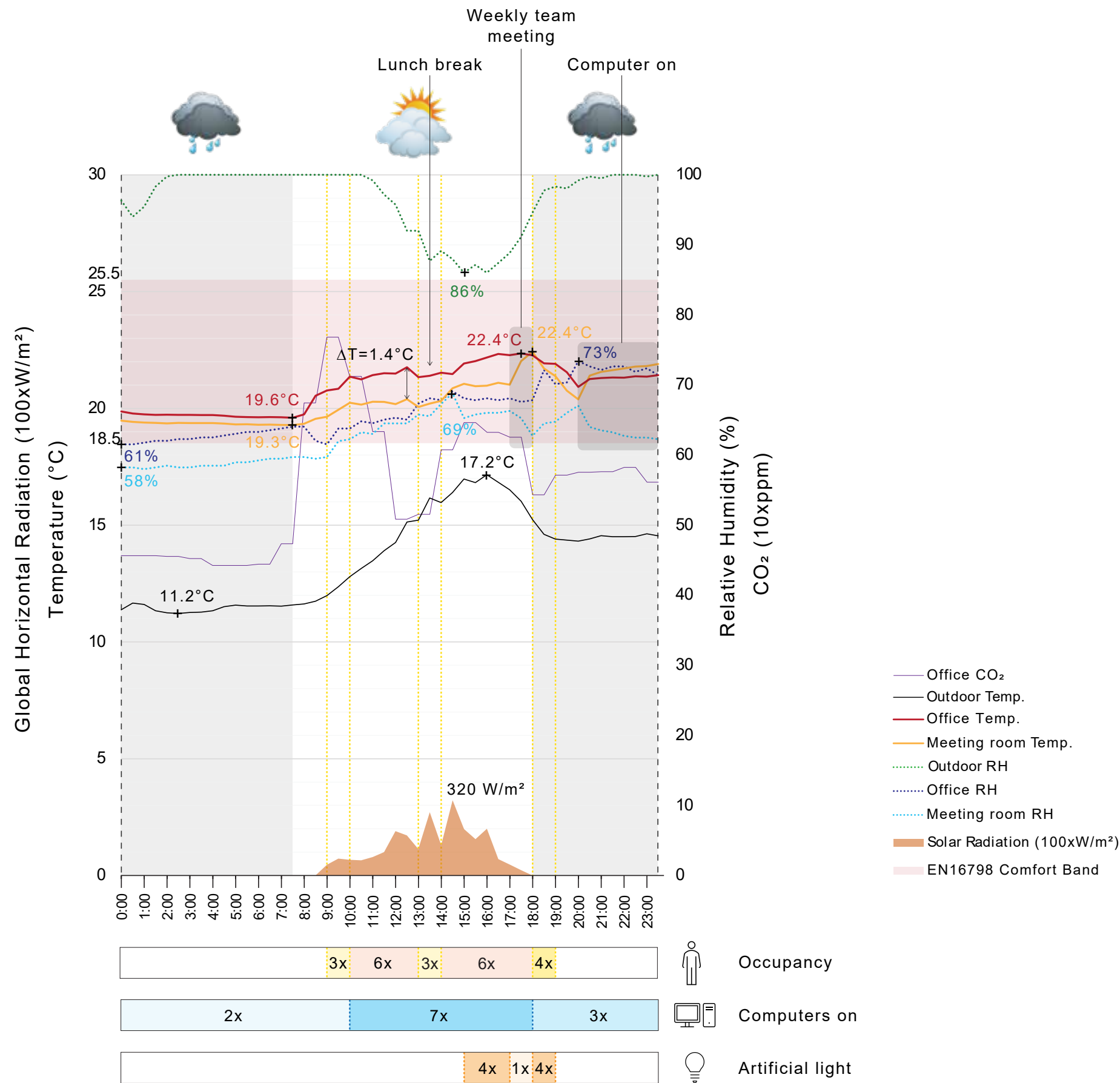


Fig. 45

Graph illustrating data obtained from data loggers and Bow's weather station for 15 October 2024.

DAY GRAPH

For a more in-depth analysis, data from Tuesday, 15 October, was plotted separately. This provides an opportunity to observe the occupancy and equipment schedule and their effects on internal temperature. The CO₂ concentration data also helps to identify occupancy patterns and moments when the window was opened. It is clear that occupants arrive between 8:00 and 9:00 AM, leave for a lunch break between 12:00 PM and 2:00 PM, and depart at 6:00 PM. Additionally, the weekly team meeting in the meeting room is noticeable from 5:00 to 6:00 PM.

Regarding equipment usage, there is a small fridge in the kitchen area, while other appliances, such as the microwave and kettle, are rarely used. Six people work in the office, and their computers are typically on during working hours. The computer in the meeting room is used for meetings and simulations, so its usage schedule is inconsistent. Additionally, two computers in the main office are often left on overnight to run simulations.

Artificial lighting does not follow a fixed schedule; it is turned on by occupants when needed, typically around 3 PM in October, although this timing varies with the seasons. The lighting system is divided into two circuits: one controls the two luminaires on the right, and the other controls the two on the left. Employees generally turn on one circuit first and add the second only when additional light is needed. Moreover, occupants at the rear tables often request lighting earlier than others. Following the construction of new buildings in front of the office, which block natural light, artificial lighting is now turned on earlier than before. Finally, the office still uses 58W fluorescent light tubes.

Temperature key data:

Outdoor

ΔT = 6 °C

Avg. = 13.6 °C

max. = 17.2 °C

min. = 11.2 °C

Main office

ΔT = 3 °C

Avg. = 20.9 °C

max. = 22.4 °C

min. = 19.6 °C

Meeting room

ΔT = 3 °C

Avg. = 20.3 °C

max. = 22.4 °C

min. = 19.3 °C

Spatial Daylight Autonomy (sDA)

% area with $\geq 50\%$ occupied hours ≥ 300 lux

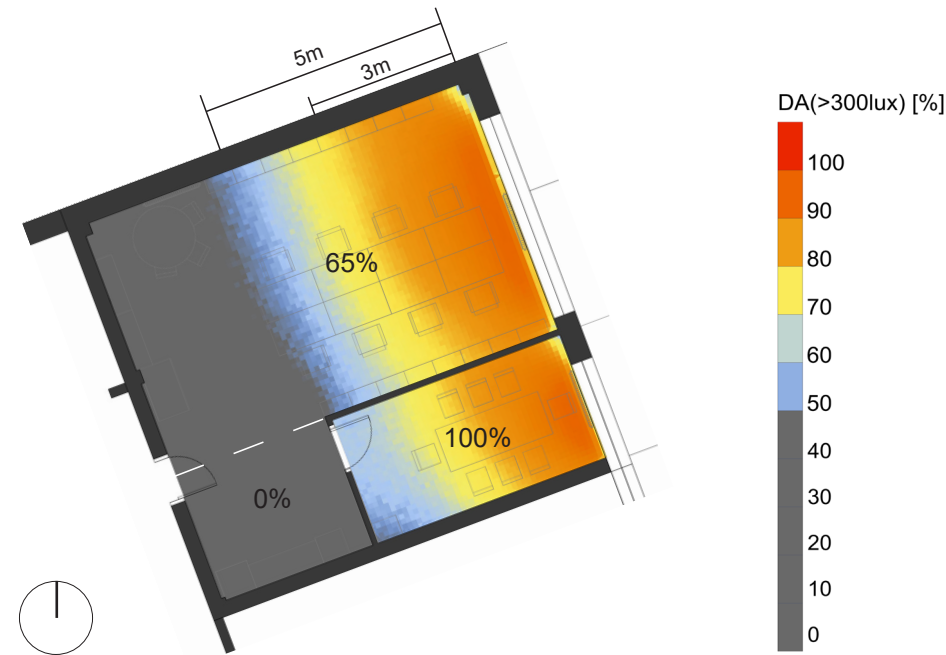


Fig. 46

Atmos Lab plan with the Spatial Daylight Autonomy results.

Spatial Useful Daylight Illuminance (sUDI)

% area with $\geq 50\%$ occupied hours $100 > 3000$ lux

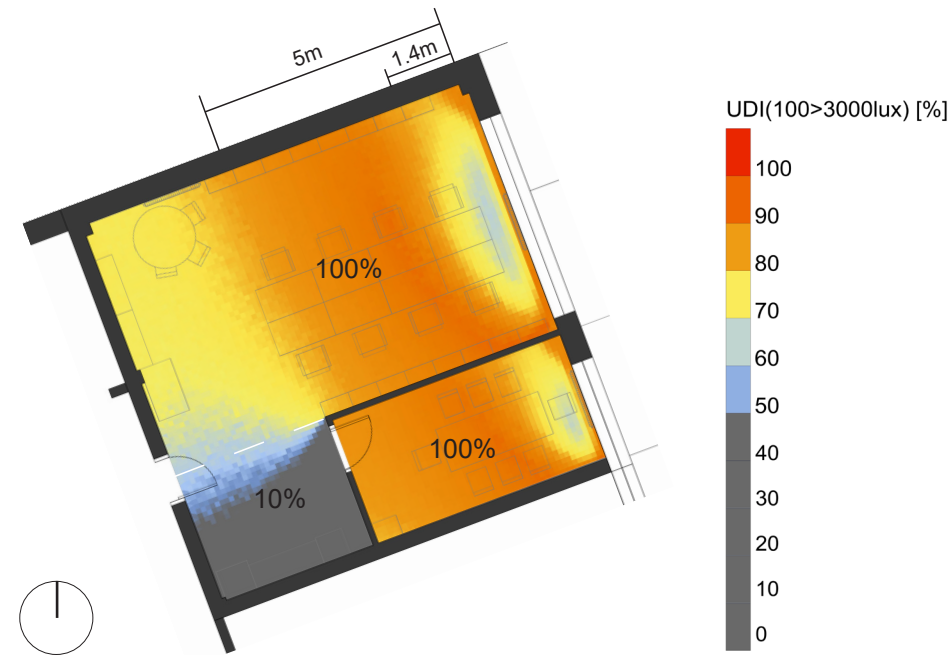


Fig. 47

Atmos Lab plan with the Spatial Useful Daylight Illuminance results.

4. ANALYTICAL INVESTIGATIONS

Analytical investigations were carried out to further explore the key findings from the site observations and interviews. The tools used by the group included the 3D modeling software Rhinoceros, along with the Ladybug and Honeybee plugins for Grasshopper for daylight, glare, and radiation simulations; Autodesk CFD for wind simulations; Optivent for ventilation calculations; and MInT for thermal calculations.

4.1. Daylight Analysis

To understand the daylight distribution in the Atmos Lab throughout the year, two simulations were conducted: Spatial Daylight Autonomy (sDA) to determine when artificial light is needed, and Spatial Useful Daylight Illuminance (sUDI) to assess the quality of daylight.

Figure 46 presents the sDA results, confirming the occupants' observations that there is insufficient daylight in the back of the office, with the two rear tables being the most disadvantaged. A sDA of at least 50% is achieved up to 5 meters from the window, and at least 80% within 3 meters of the window. 65% of the main office area receives at least 300 lux for 50% of the occupied hours (8 AM to 6 PM), which exceeds the WELL Building Standard threshold of 55% for sDA. The meeting room, 4.85 meters deep, has excellent daylight distribution, with 100% of the space receiving at least 300 lux for 50% of the occupied hours. Although the kitchen area never reaches 300 lux with natural daylight, it is not an area of prolonged occupation and can rely on artificial lighting when used.

Figure 47 shows the sUDI results, indicating that good-quality daylight is available between 1.4 and 5 meters from the window, with excessive daylight (>3000 lux) near the windows and insufficient daylight (<100 lux) at the back of the office. Both the main office and the meeting room have 100% of the area with at least 100 and up to 3000 lux for 50% of the occupied hours.

Finally, point-in-time illuminance simulations were performed to further explore the daylight performance of the office during different seasons and times of day. The simulation results, presented in Figure 48, show that when the sky is overcast and only indirect daylight is available, there is no significant difference between morning and afternoon results. However, during summer, when the sky is intermediate, significantly more daylight enters the space in the morning, since the window is oriented to the northeast. Additionally, daylight issues are more pronounced in winter, when less daylight is available outdoors, making it difficult to achieve 300 lux inside the space.

The reflectances used in the simulations were: concrete floor 30%, white walls 75%, white ceiling 80%, brick external wall 30%, black external wall 5%, metal roof 60%, context 30%, ground 20% (<https://spectraldb.com>), and glass transmittance 70%. The analysis grid was set to a height of 75cm from the floor.

Point-in-Time

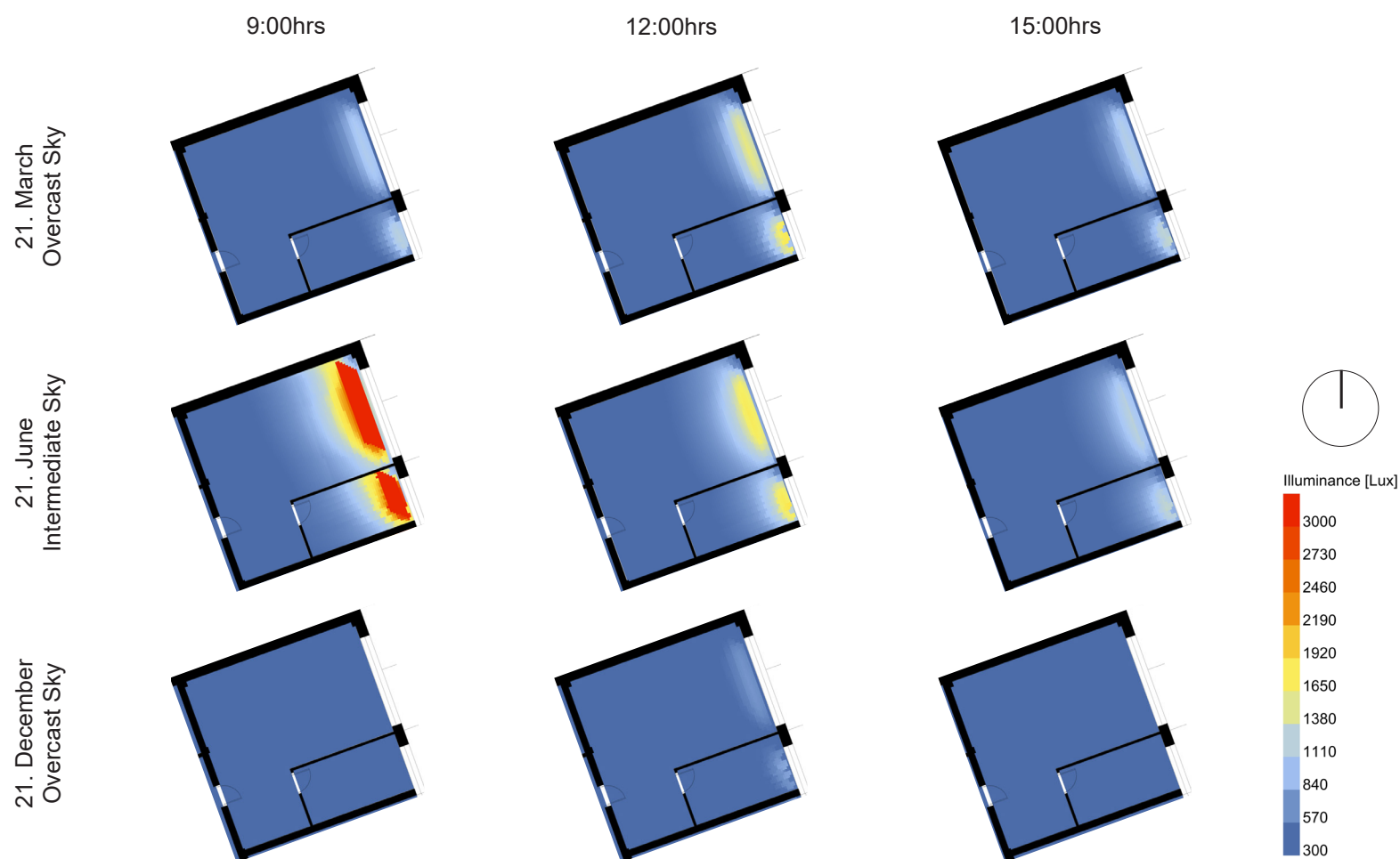


Fig. 48

Atmos Lab plans with point-in-time illuminance results.

4.2. Glare Studies

During the interviews, the occupants noted that the users of the two desks closer to the window experience some glare. The occupant facing southeast (Florenzia) encounters glare during summer mornings, while the occupant facing northeast (Olivier) is beginning to experience glare in the afternoon, primarily due to sunlight being reflected by the reflective façade element of the new buildings.

The Daylight Glare Probability (DGP) false-color images (Figures 49 and 50) illustrate the likelihood of discomfort glare at specific times of the day and year. From Florenzia's desk, disturbing glare is observed from both the sky and the metal roof of the ground floor (21 June, 9 AM). From Olivier's desk, perceptible glare originates from the reflective façade of the new building (21 September, 3 PM).

The Glare Autonomy (GA) simulation, shown in Figure 51, quantifies the percentage of time when specific points in the space are free from glare. This simulation confirms the likelihood of discomfort glare at both Florenzia's and Olivier's desks when facing the window. It also suggests that moving less than a meter into the interior of the office would solve the glare issue. Additionally, during the summer months, closing the curtain would serve as another practical solution to reduce glare.

Daylight Glare Probability (DGP)
Florenzia's desk

Sunny sky
June 21st
9:00 am

DGP 0.42
Disturbing

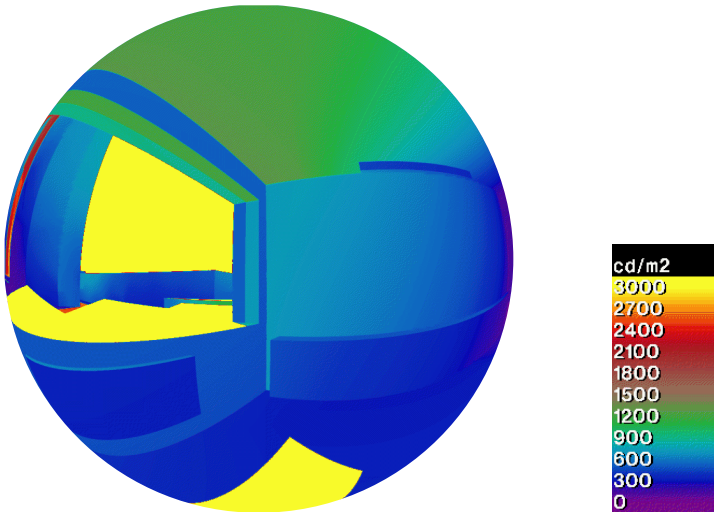


Fig. 49
Falsecolour image from Florenzia's desk and looking to the window.

Daylight Glare Probability (DGP)
Olivier's desk

Sunny sky
September 21st
3:00 pm

DGP 0.30
Perceptible

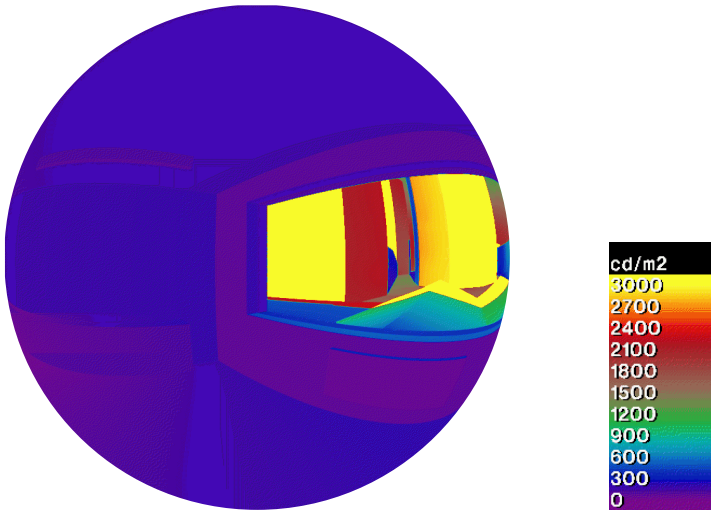


Fig. 50
Falsecolour image from Olivier's desk and looking to the window.

Glare Autonomy (GA)
% of time without glare

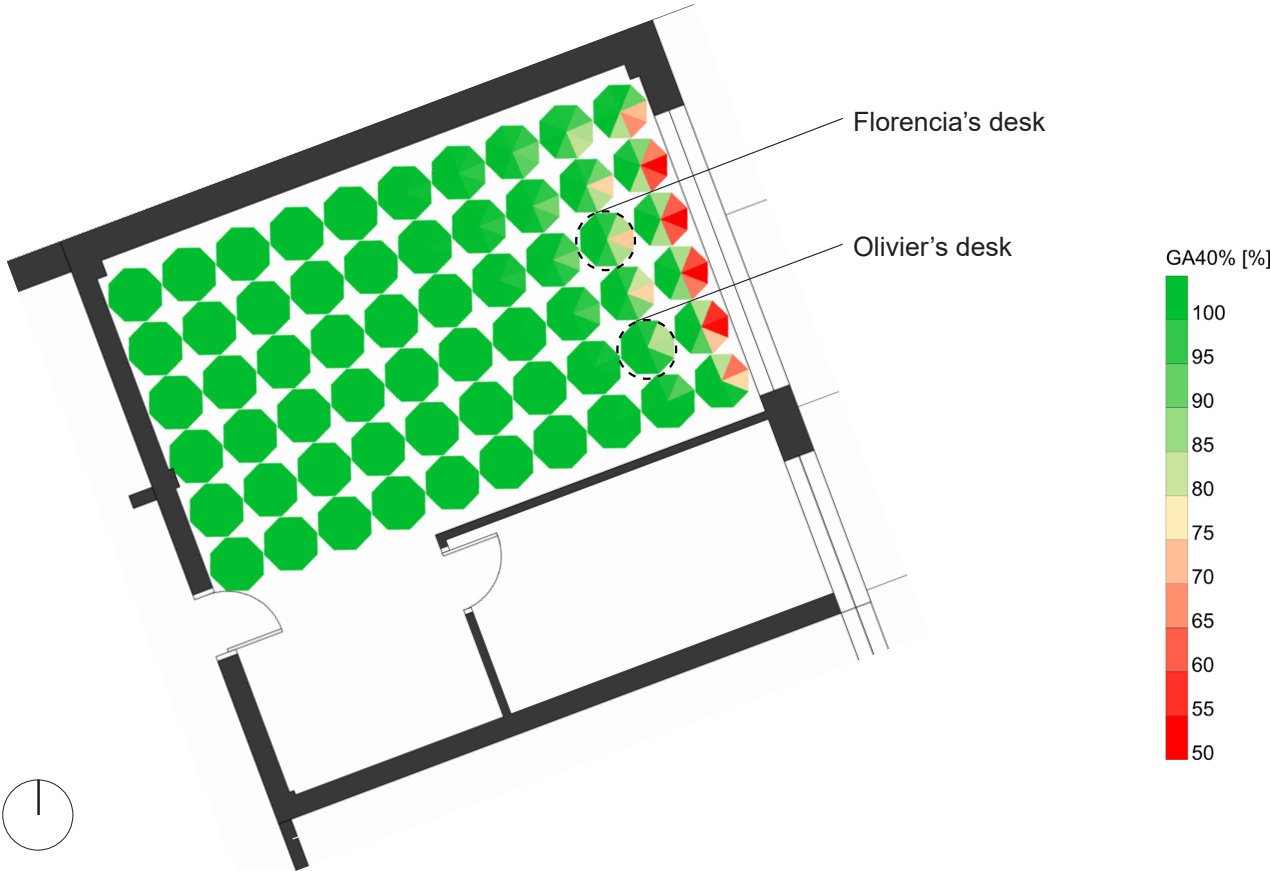


Fig. 51
Glare Autonomy simulation results.

Scenarios for a typical summer day with Optivent:

1. Single Sided Ventilation:

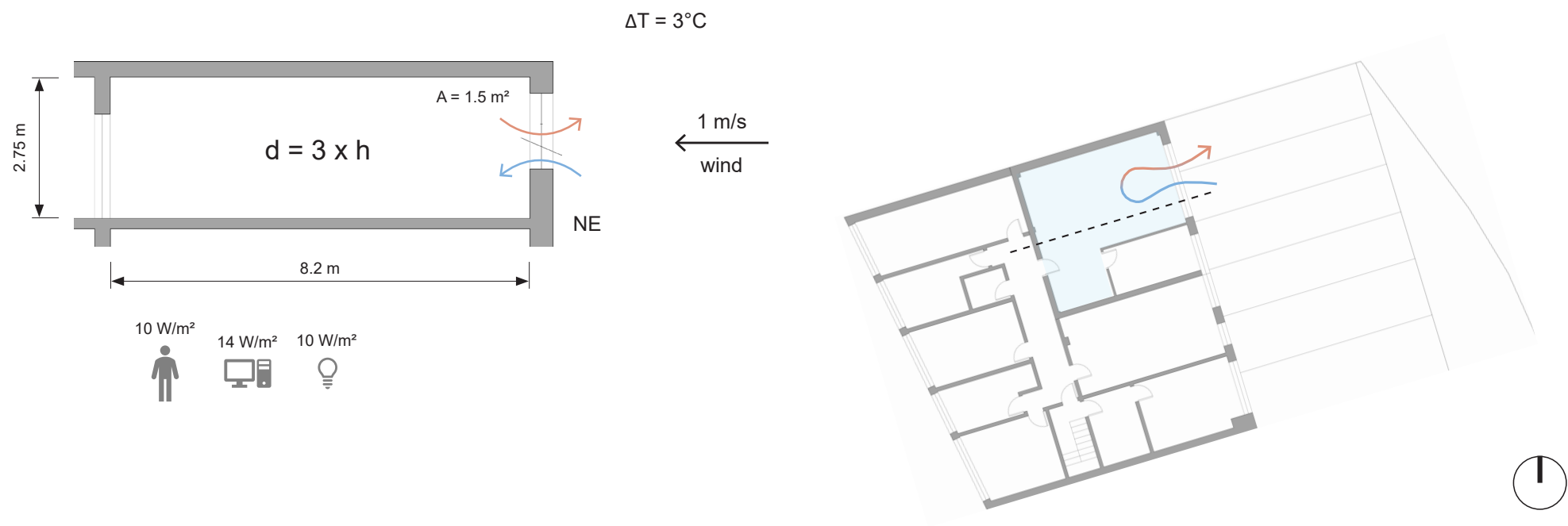


Fig. 52
Section and plan diagrams of the single sided ventilation scenario for the Atmos Lab Office.

2. Cross Ventilation:

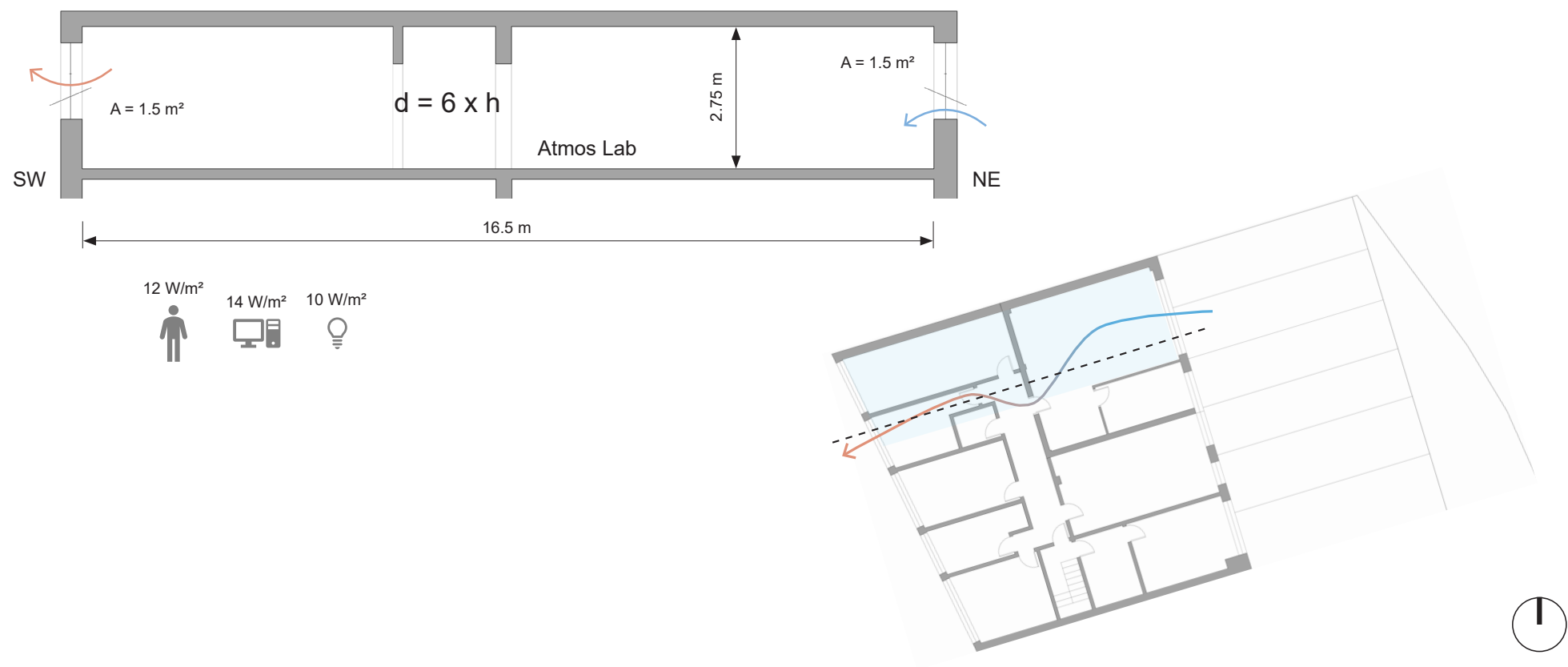


Fig. 53
Section and plan diagrams of the cross ventilation scenario for the Atmos Lab Office.

4.3. Ventilation Analysis

A limitation in achieving proper ventilation during the summer was identified; therefore, two ventilation scenarios were analyzed using Optivent. The first scenario assumes the corridor door is closed, allowing only single-sided ventilation, as shown in Fig. 52. The second scenario assumes that all the offices in the building have their corridor doors open, enabling cross ventilation (Fig. 53).

Both scenarios were calculated for a typical summer day, 21 June, at midday, with a temperature difference of 3°C between the interior and exterior, a wind speed of 1 m/s, and internal heat gains of 14 W/m² from equipment, 10 W/m² from artificial lighting, and 10 or 12 W/m² from occupants. These values were calculated based on Atmos Lab's internal loads and adapted for the adjacent office.

The graph in Fig. 54 shows that neither scenario achieves the required air changes per hour (ACH) for cooling. However, there is a significant increase in the ventilation rate when comparing single-sided to cross ventilation, from 3.6 to 8 ACH.

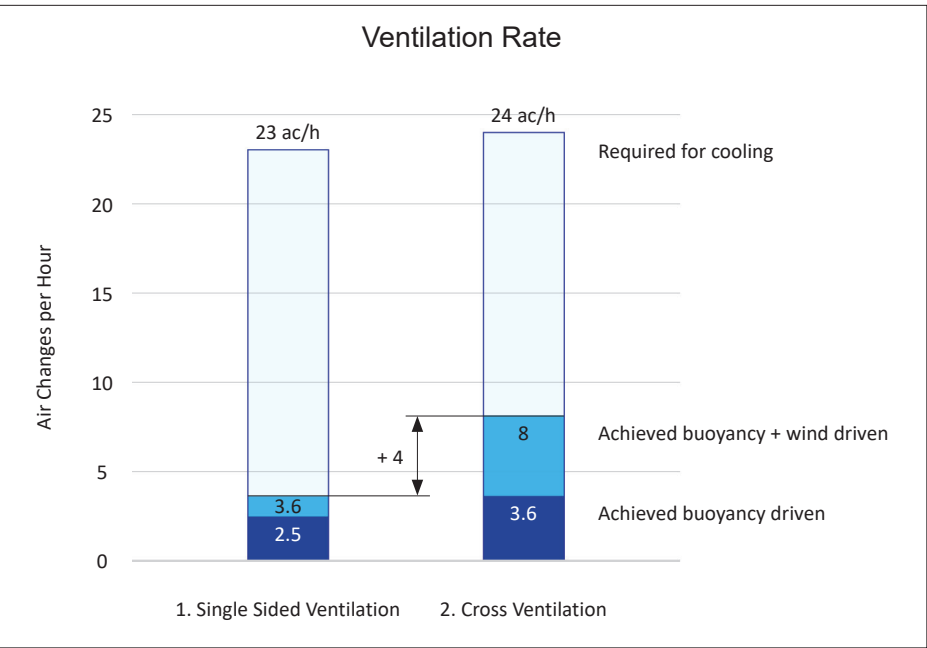


Fig. 54
Graph showing the air changes required for cooling and achieved with single sided and cross ventilation scenarios.

4.4. Wind Studies

Since wind-driven ventilation is crucial for removing internal heat generated in the office during the warm season, a Computational Fluid Dynamics (CFD) simulation was conducted. The wind profile used in the simulation was derived from wind velocity and direction data in the London St. James EPW file, considering only the occupied hours between June and August (Fig. 54). The predominant wind direction during this period is southwest (Fig. 55).

The CFD simulation results are presented in Fig. 56, where two vortices can be observed in the section, highlighting the influence of the new buildings on wind behavior at the back of the Atmos Lab building. Under these conditions, the wind speed on both façades is 0.5 m/s, with a pressure difference of 0.3 Pa, where the front façade serves as the inlet and the back façade as the outlet. However, a pressure difference of 0.3 Pa may be insufficient to drive effective cross ventilation, emphasizing the importance of well-designed openings to ensure adequate ventilation rates during the summer and prevent overheating.

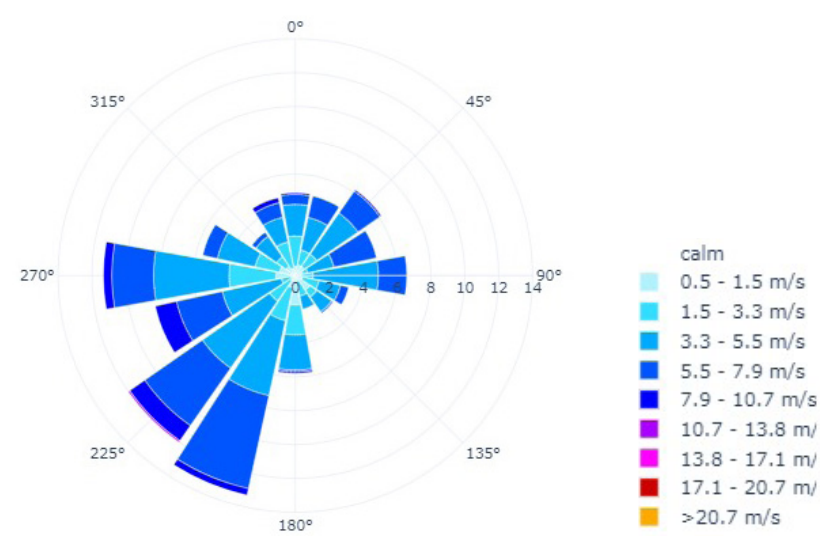


Fig. 54
Wind direction and velocity during occupied hours in the warm season (Source: Clima Tool).

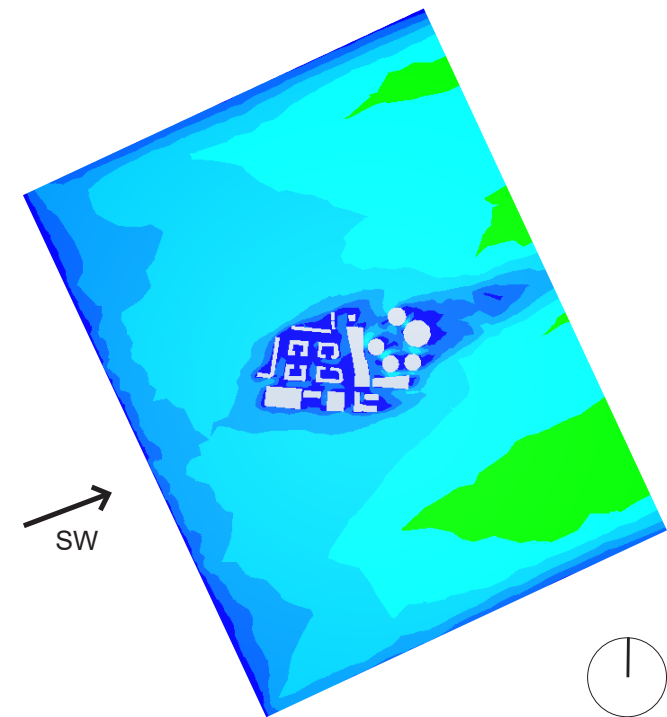


Fig. 55
Image of the wind tunnel and wind direction used for CFD simulations.

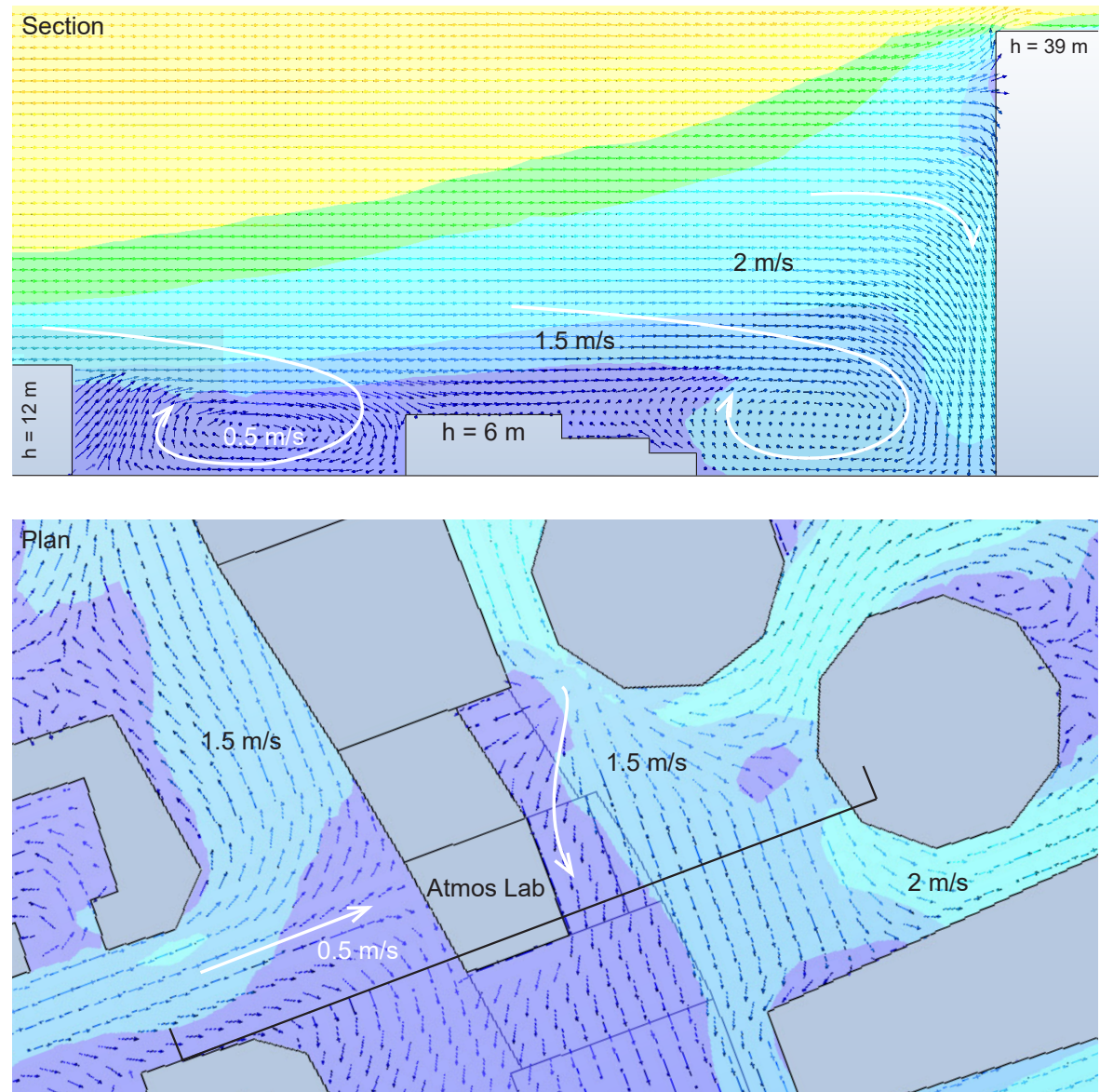


Fig. 56
CCFD simulation results in plan and section for the warm season wind profile and predominant wind direction (SW).

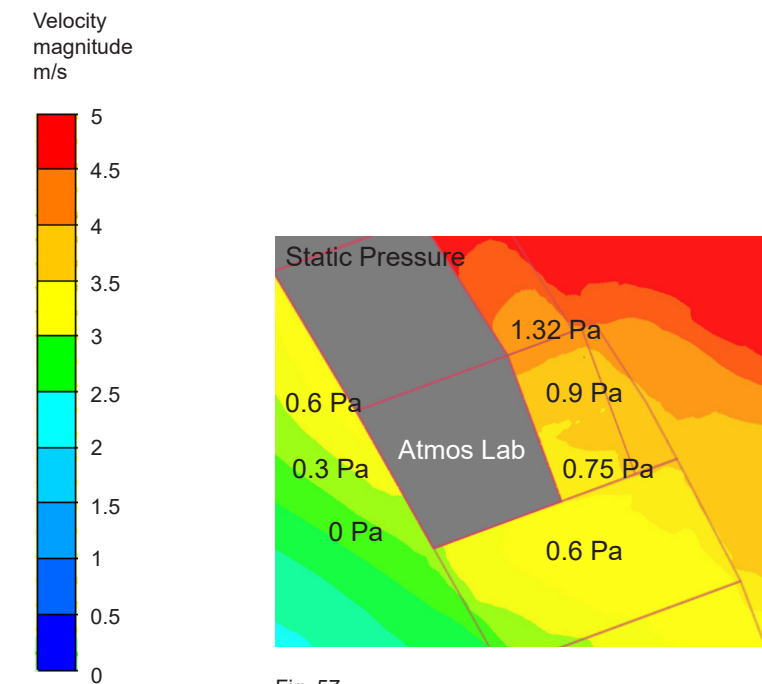


Fig. 57
Static pressure distribution on the Atmos Lab office building façades.

4.5. Shade Study

Figure 58 illustrates the shading pattern of the Atmos Lab office building. Throughout the year, the Atmos Lab façade receives direct sunlight in the mornings and remains shaded from midday onward. The roof is rarely shaded, except during winter when the sun is lower in the sky. This pattern indicates that the façade primarily receives sunlight during cooler morning hours, while the insulated roof minimizes the impact of solar radiation, even during prolonged exposure.

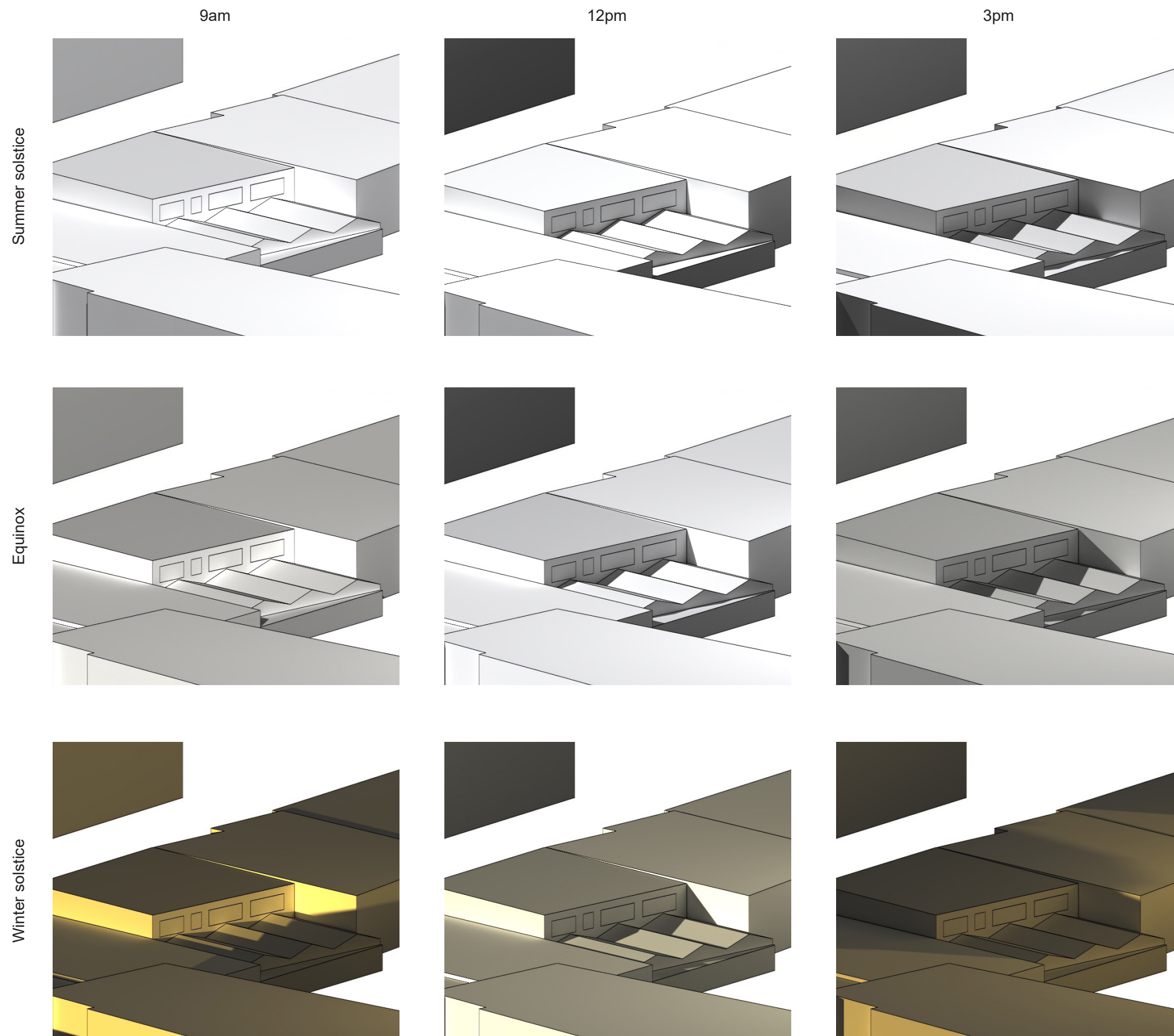


Fig. 58
Shade study of Atmos Lab Office façade.

4.6. Solar Radiation

To analyze the solar radiation incident on the Atmos Lab office windows, virtual typical summer and winter days were generated by averaging data from the respective typical summer and winter weeks of the London St. James EPW file. These virtual days are shown in Figs. 59 and 60.

The cumulative incident radiation on the Atmos Lab windows and the average air temperature during occupied hours were calculated for use in the MInT calculator as part of the thermal analysis.

Figures 61 to 64 show that the building's front façade receives more than twice the solar radiation compared to the back façade. This is because the back façade only receives direct sunlight in the morning, while the front façade is exposed to it in the afternoon. Additionally, it is clear that significantly less solar radiation reaches the building's windows during winter than in summer.

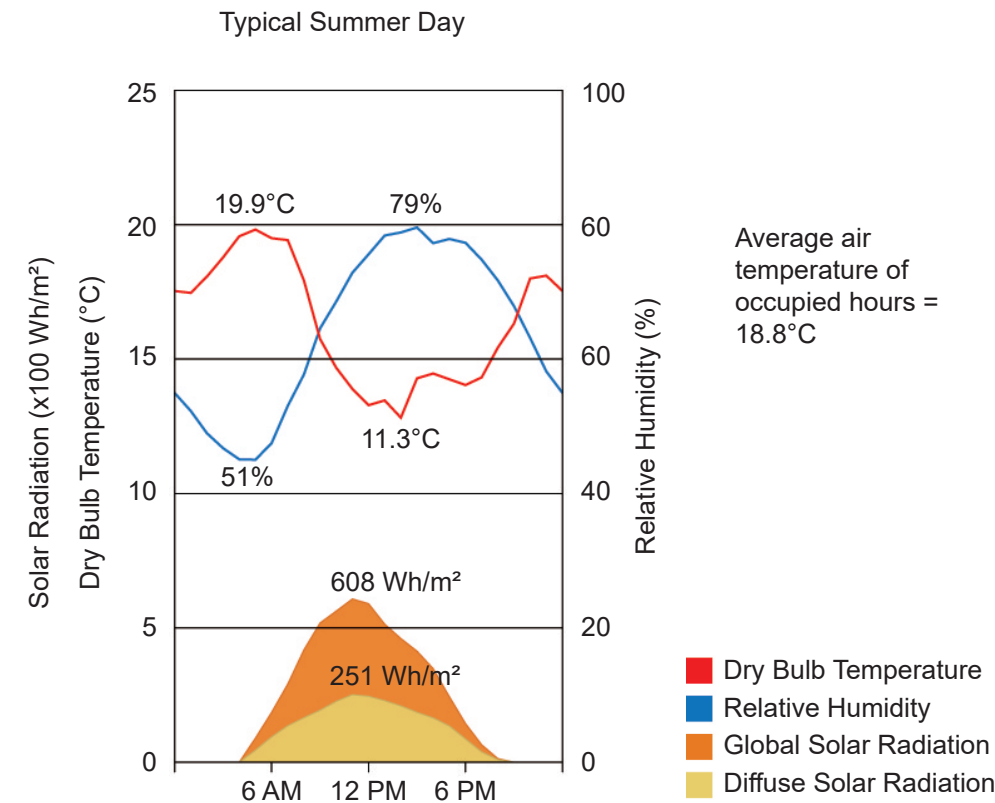


Fig. 59
Graph of the typical summer day for London.

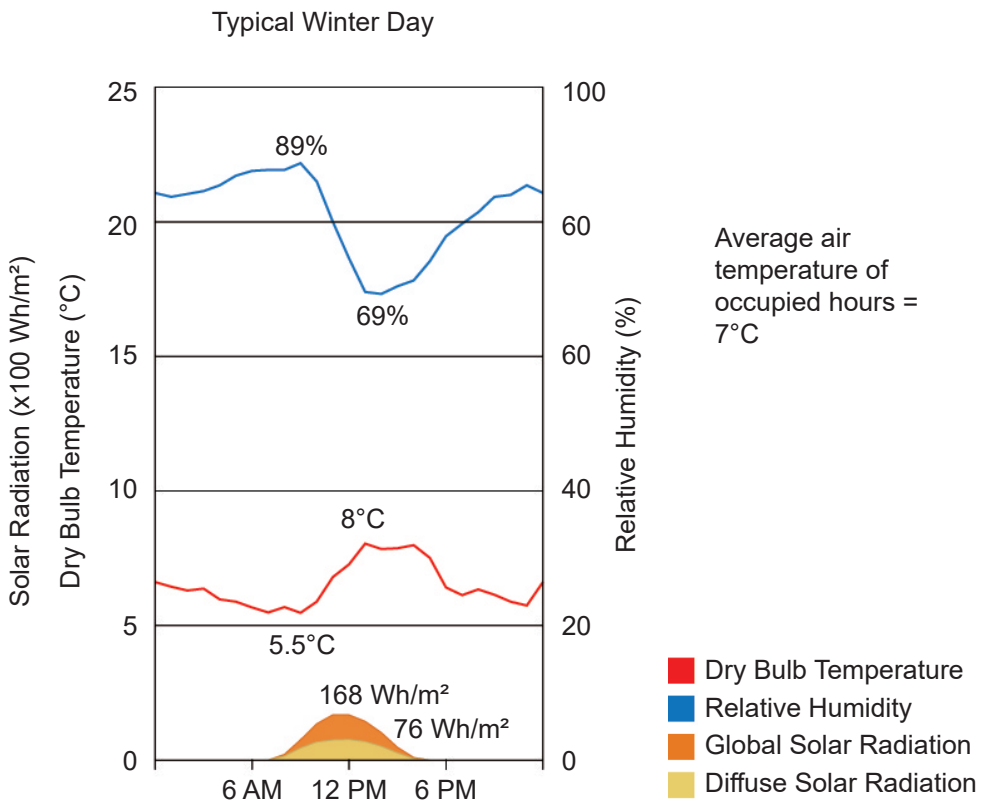


Fig. 60
Graph of the typical winter day for London.

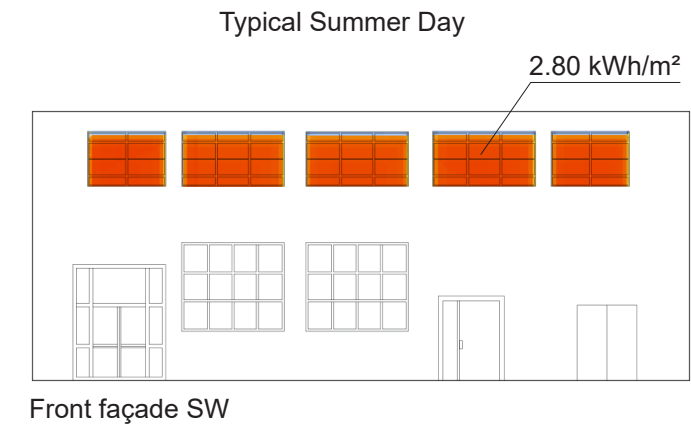


Fig. 61
Cumulative Incident radiation on the SW façade windows for a typical summer day.

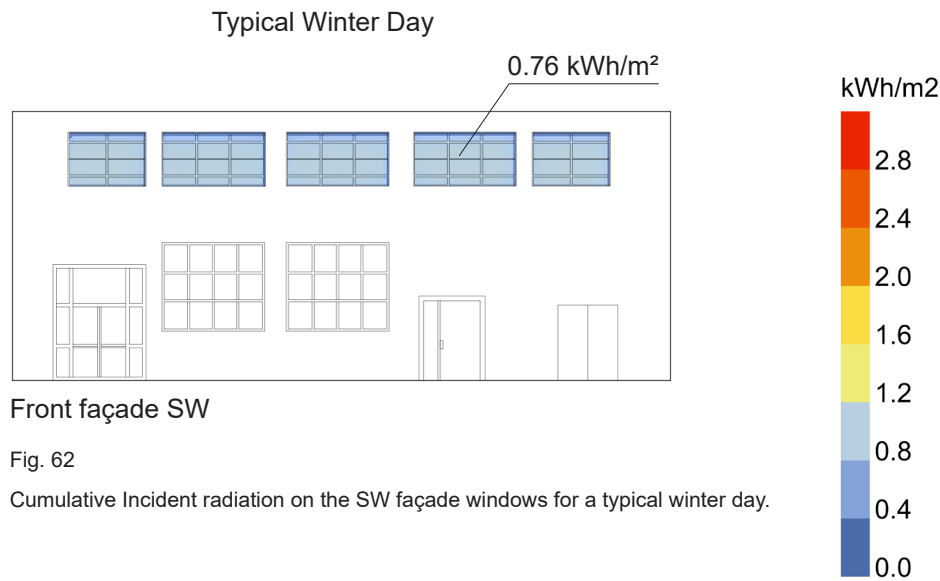


Fig. 62
Cumulative Incident radiation on the SW façade windows for a typical winter day.

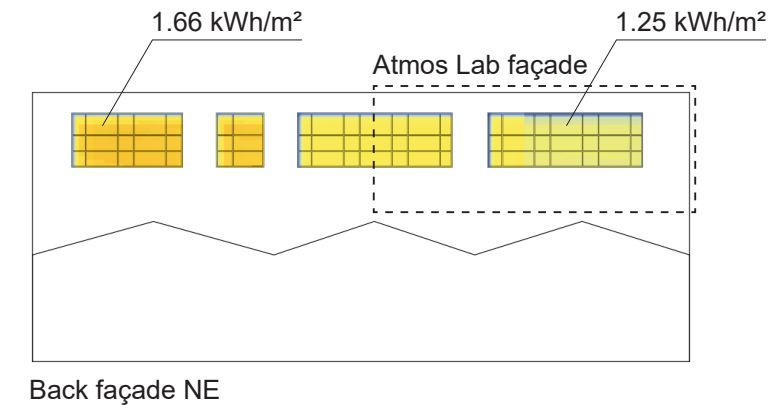


Fig. 63
Cumulative Incident radiation on the NE façade windows for a typical summer day.

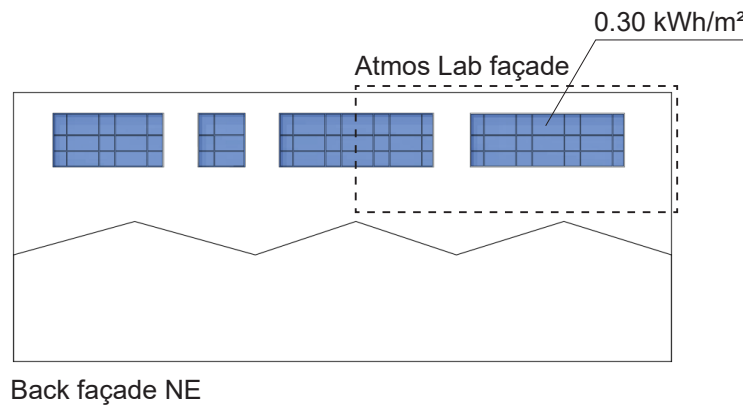


Fig. 64
Cumulative Incident radiation on the NE façade windows for a typical winter day.

4.7. Thermal Analysis

YEAR DATA

The Atmos Lab Office records the internal temperature of the main office area, allowing us to analyze how the temperature fluctuates throughout the year. Notably, the temperature rarely drops below 19°C, except during the Christmas and New Year break. This suggests that, despite Atmos Lab turning off its radiator, the heating from adjacent offices keeps the temperature above 19°C. The external temperature data, plotted in the graph, was sourced from the London Limehouse station on the Wunderground website for the corresponding period.

By comparing internal and external temperatures, we identified the period when the building's central heating is active, which spans from November to February. Moreover, the internal temperature generally remains within the EN comfort band. However, some overheating

periods are noticeable, especially during the warmer months. The temperature exceeding the comfort band accounts for less than 1% of the occupied hours (9 AM to 7 PM) throughout the year, but represents 2.2% of the occupied hours during the warm period. Nevertheless, this does not indicate a significant overheating problem.

CO₂ concentration levels provide further insight into the building's natural ventilation. It is evident that CO₂ values are consistently lower during the warm months, due to higher ventilation rates. CO₂ levels remain below 2,000 ppm, the threshold for moderate air quality, and are typically below 1,000 ppm, which is considered optimal for good air quality. This observation aligns with feedback from interviews, where it was reported that CO₂ levels are regularly monitored, and windows are slightly opened to introduce fresh air when CO₂ levels are high, even during colder periods.

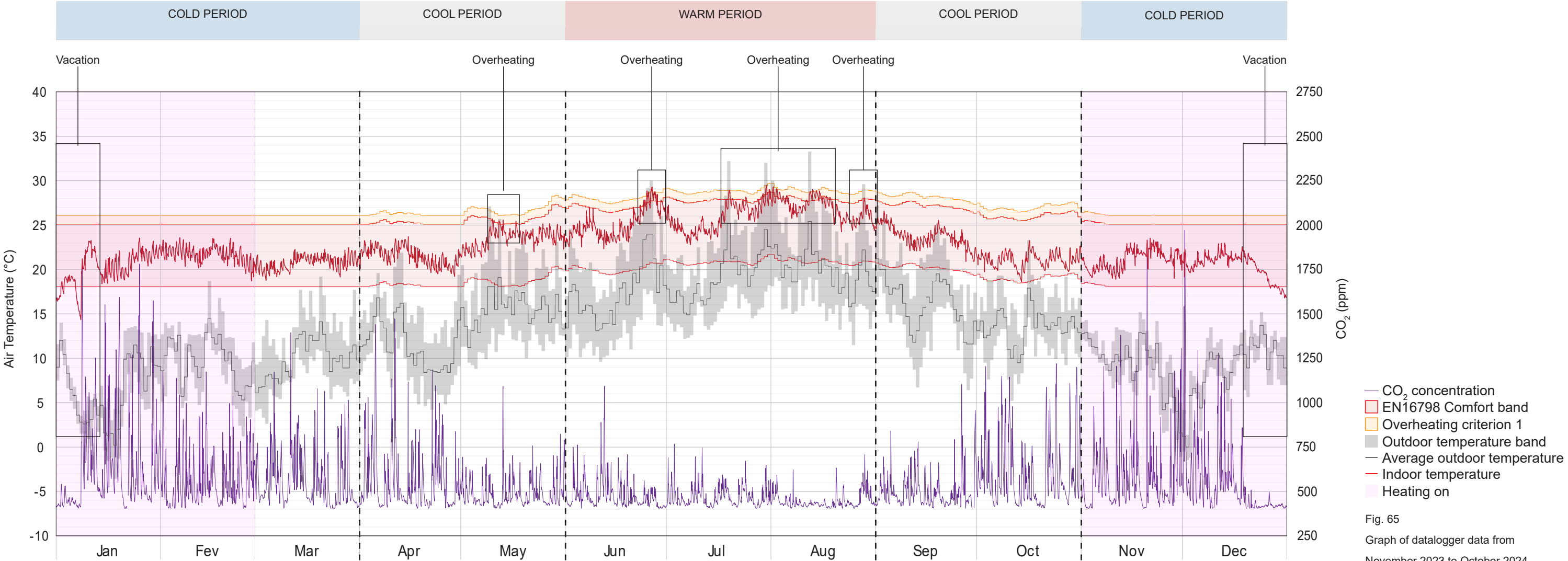


Fig. 65
Graph of datalogger data from
November 2023 to October 2024.

HEAT LOSS COEFFICIENT

To assess the performance of the office envelope, the U-value and heat loss coefficient of the external walls, windows, and roof are presented in Fig. 66. Among these elements, the windows represent the primary source of heat loss due to the single glazing. When comparing the heat loss coefficient of Atmos Lab to other building regulations, it is evident that the office is twice as well insulated as the building regulations of 1972 would require. However, it is also approximately twice as poorly insulated as the current standards set by Part L (Fig. 67).

The diurnal thermal storage (DHS) analysis indicates that the building has significant thermal mass, primarily due to the exposed concrete floor, roof, and brick walls (Fig. 68). While this thermal mass plays a crucial role in moderating temperature fluctuations throughout the day, its potential is not fully utilized during summer due to insufficient ventilation.

ENERGY BALANCE

The energy balance graph (Fig. 69) for typical summer and winter days shows that appliances are the main internal sources of heat, with the windows again emerging as a notable point of concern in terms of heat loss.

Heat Loss Coefficient:

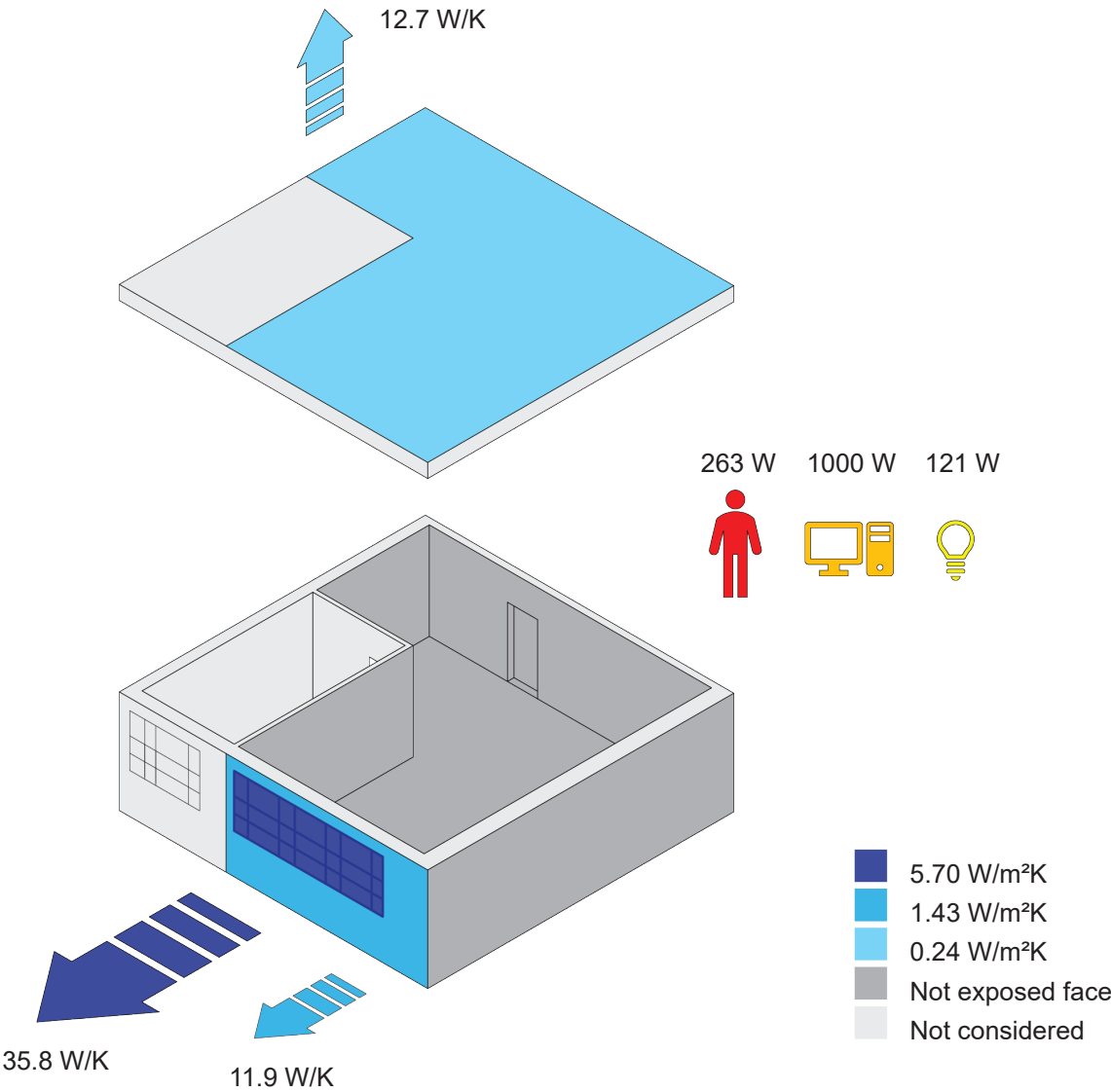


Fig. 66
Diagram of the U-value and Heat Loss Coefficient of the Atmos Lab envelope.

Heat Loss Coefficient Scale:

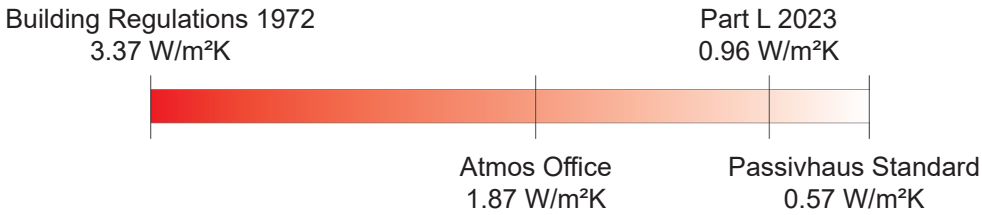


Fig. 67
Heat Loss Coefficient scale.

Energy Balance:

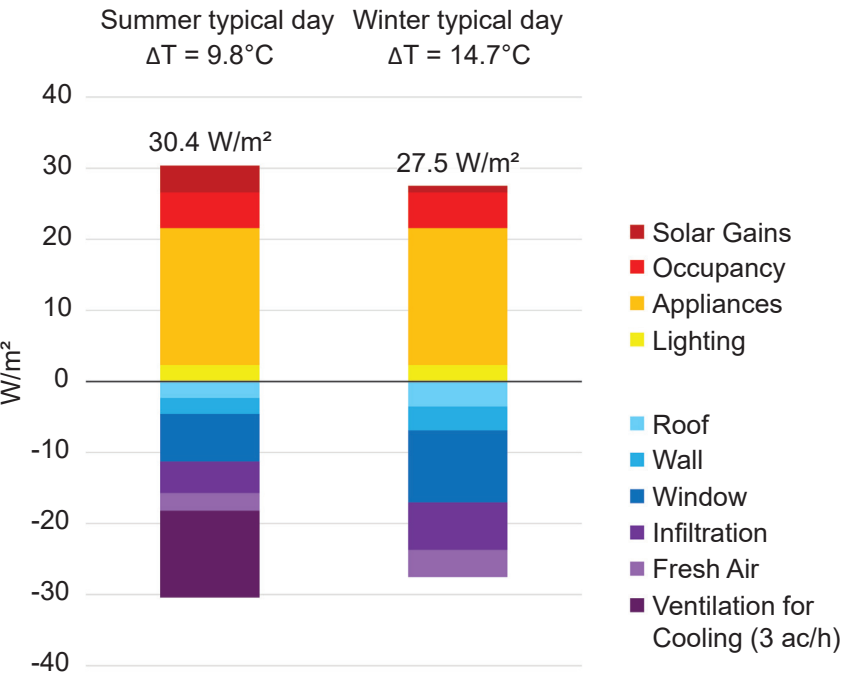


Fig. 69
Graph of the energy balance of Atmos Lab for typical summer and winter days.

Diurnal Heat Storage (DHS):

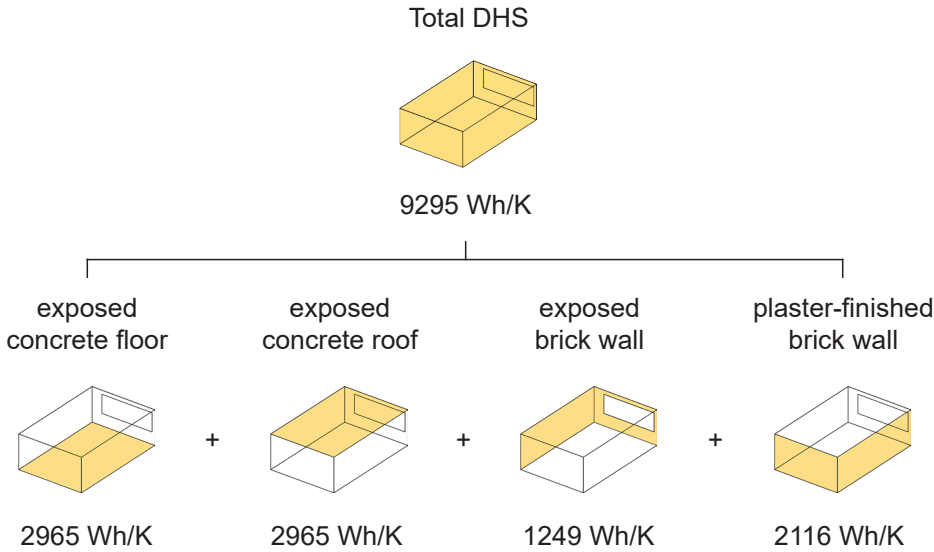


Fig. 68
Diurnal heat storage of Atmos Lab Office.

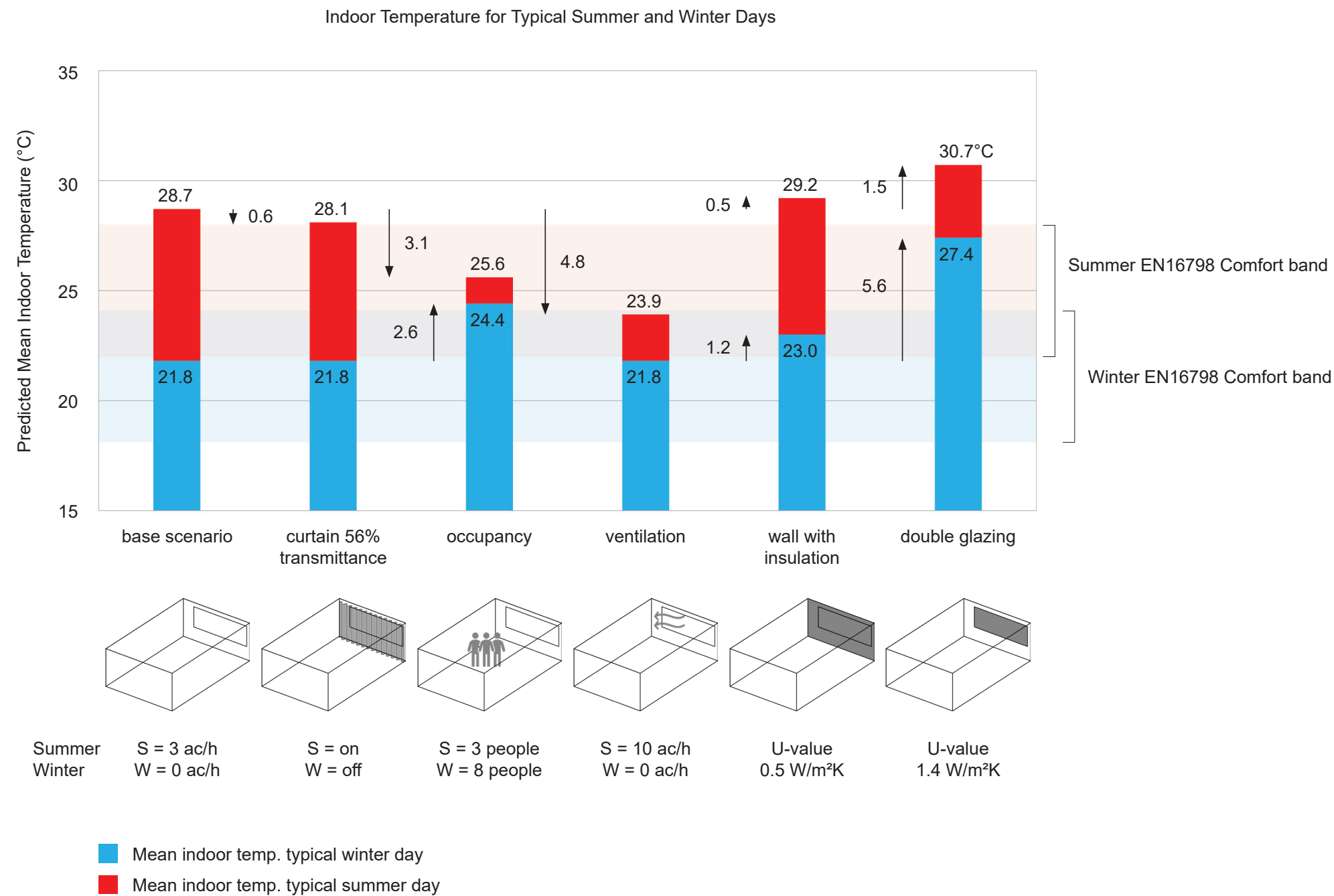
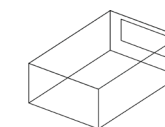


Fig. 70
Graph of the internal temperature fluctuations due to different scenarios.

THERMAL ITERATIONS

Several iterations using the MInT tool were conducted to evaluate the effects of various scenarios on internal temperature. These scenarios included: the base case, closing the curtains during summer, adjusting occupancy by increasing the number of people in winter and reducing it in summer, increasing the ventilation rate to 10 ACH during summer, adding insulation to the external walls, and upgrading the windows to double glazing. The results of these simulations highlighted that increasing ventilation is the most effective cooling strategy, while replacing the windows proved to be the most effective solution for managing winter conditions. Although the internal temperature occasionally exceeds the comfort band during winter, it is important to note that window infiltration, identified as a major issue during interviews, significantly contributes to these fluctuations. To provide occupants with better control over their environment, improving window performance would be an effective strategy.

base scenario inputs



Roof U-value	0.24 W/m²K
Windows U-value	5.7 W/m²K
External wall U-value	1.43 W/m²K

Infiltration rate	0.5 ac/h
Fresh air	30m³/person hr
Vent. for cooling	S = 3 / W = 0 ac/h

Occupancy	263 W (6 x 117 W for 9h)
Fridge	100 W (1 x 100 W for 24h)
Computers	900 W (6 x 400W for 9h)
Lights	121 W (10 x 58 W for 5h)

Incident solar	S = 1.25 / W = 0.3 kWh/m² per day (80% transmitted)
Outdoor temp.	S = 18.8 / W = 7 °C

5.1 Ventilation Analysis

From a ventilation perspective, replacing the window is crucial to create more operable areas and improve airflow. Adding a skylight at the back of the office is essential for facilitating cross ventilation, as it is important for the space to have its own independent ventilation system, rather than relying on adjacent offices.

In Fig. 72, the proposed skylight and new window are sized to enable effective cross ventilation. First, we considered 90% of the window area as the effective aperture, resulting in 5.4m². We then tested various effective opening areas for the skylight to achieve the necessary air changes per hour (ACH) to cool the space on a typical summer day, settling on 1.35m².

In the ventilation rate graph (Fig. 73), the required ACH for cooling is compared with the achieved buoyancy-driven and buoyancy + wind-driven ventilation in each scenario. The proposal to add a new window and skylight nearly meets the required ACH, demonstrating a significant improvement in ventilation performance. The 10 ACH achieved solely through buoyancy-driven ventilation would be sufficient to provide good ventilation and night ventilation. However, when wind conditions are favorable, it would be capable of removing all internal heat gains and providing a cooling effect due to increased air speed.

PROPOSAL SCENARIO WITH OPTIVENT

NEW WINDOW + SKYLIGHT

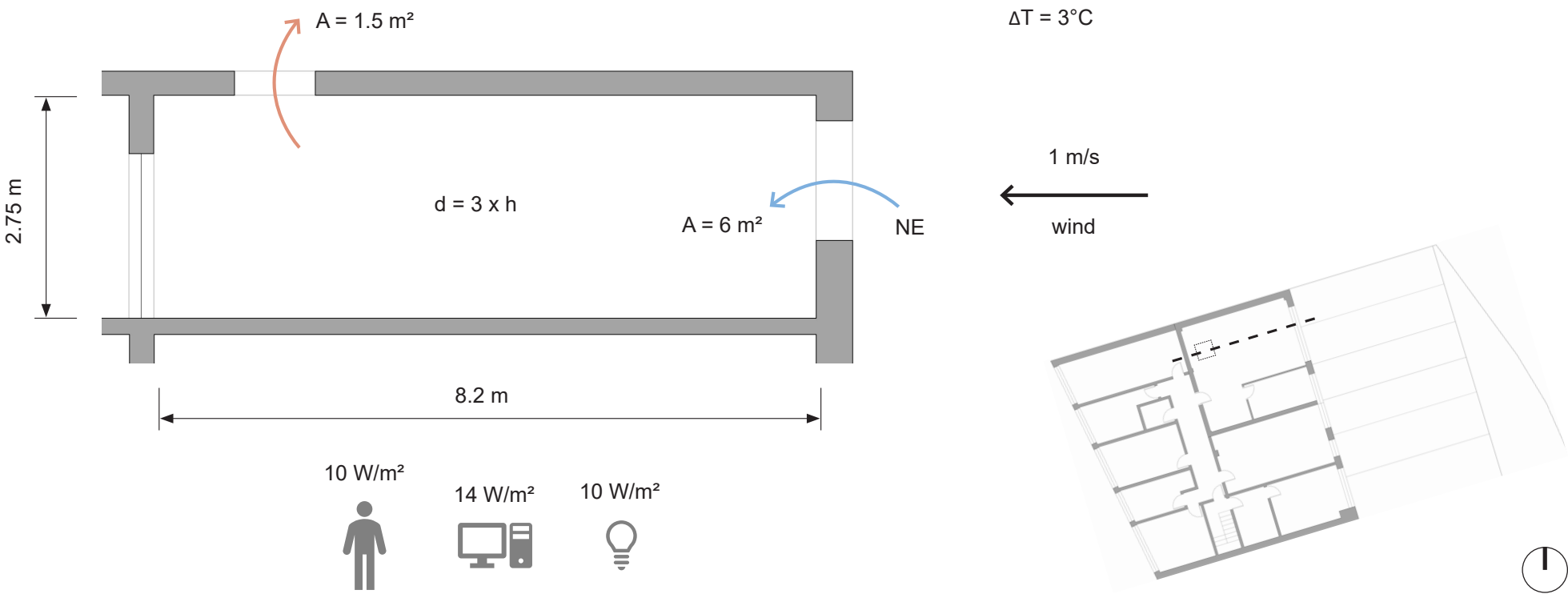


Fig. 72
Section and plan diagrams of the cross ventilation scenario proposal for the Atmos Lab Office.

VENTILATION RATE

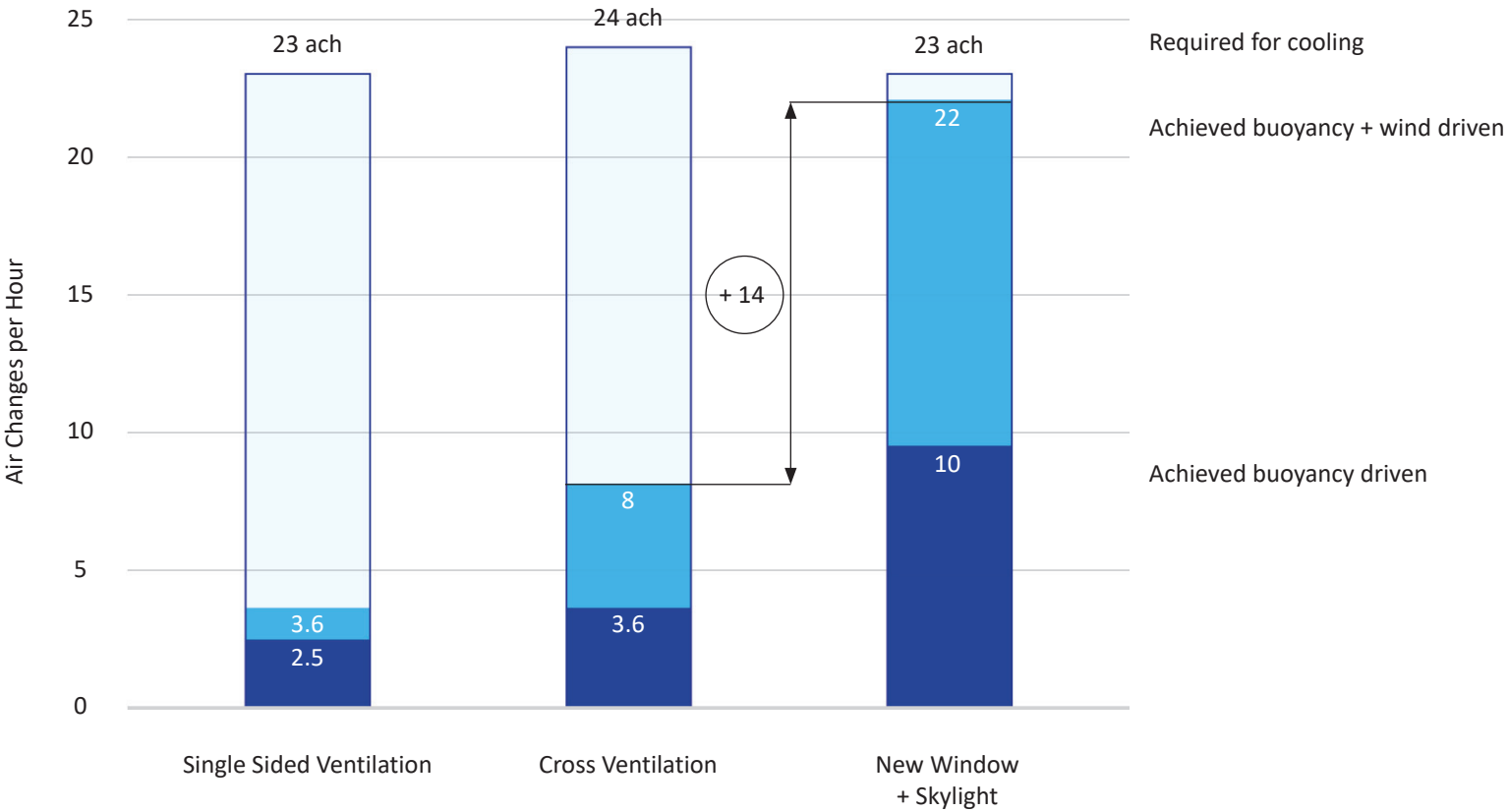


Fig. 73
Graph showing the air changes required for cooling and achieved with singled sided, cross ventilation and New window + Skylight scenarios.

INDOOR TEMPERATURE FOR TYPICAL SUMMER AND WINTER DAYS

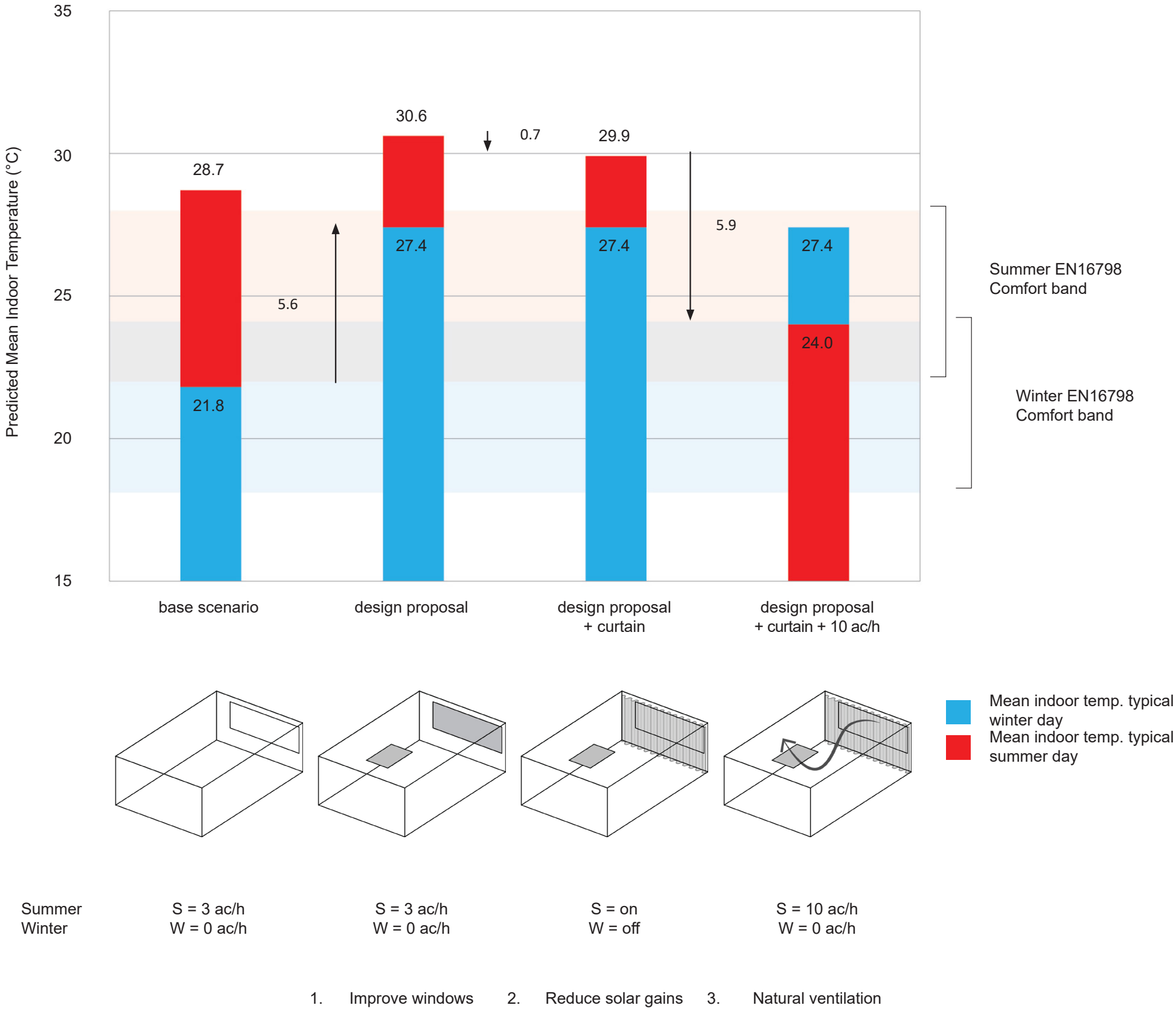


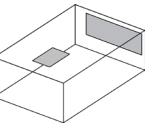
Fig. 74
Graph of the internal temperature fluctuations due to different scenarios.

5.2. Thermal Iterations

We conducted additional thermal iterations using the MInT tool to evaluate the effects of the proposed glazed areas and enhanced ventilation. We started with the base case and then incorporated the new proposal, upgrading the window to a double-glazed model. Next, we analyzed the effect of closing the curtains during the summer and, finally, the impact of increasing ventilation to 10 ACH.

The results showed significant improvements. The new proposal increased winter temperatures by 5.6°C compared to the base case and addressed the issue of infiltration through the window gaps. The use of curtains in the summer reduced temperatures by 0.7°C, while increasing ventilation to 10 ACH significantly lowered summer temperatures (5.9°C). As a result, both summer and winter temperatures were maintained within the EN16798 Comfort Band.

design proposal scenario inputs



Roof U-value	0.24 W/m²K
Windows U-value	1.4 W/m²K
External wall U-value	1.43 W/m²K
Infiltration rate	0.5 ac/h
Fresh air	30m³/person hr
Vent. for cooling	S = 3 / W = 0 ac/h
Occupancy	263 W (6 x 117 W for 9h)
Fridge	100 W (1 x 100 W for 24h)
Computers	900 W (6 x 400W for 9h)
Lights	121 W (10 x 58 W for 5h)
Incident solar	S = 1.25 / W = 0.3 kWh/m² per day (60% transmitted)
Outdoor temp.	S = 18.8 / W = 7 °C

5.3 Daylight Analysis

As with the thermal iterations, we continued with simulations for Spatial Daylight Autonomy (sDA) and Spatial Useful Daylight Illuminance (sUDI), similar to the analytical investigations, to determine the optimal position for the proposed skylight at the rear of the office. Focusing on the workspace area and using the current situation as a baseline, the following results were obtained:

One FLat Skylight:

The skylight follows the orthogonal layout of the space, positioned parallel to the access wall, and is a flat roof skylight. This configuration achieves 100% sDA in the area, providing at least 300 lux for 50% of the occupied hours. However, only 2.5 meters near the window and a circular area with a 2-meter diameter around the skylight achieve at least 80% sDA.

For sUDI, 100% of the space receives between 100 and 3000 lux for 50% of the occupied hours, including the kitchen area.

Two Flat Skylights:

In this scenario, the same skylight opening area is divided into two skylights. The sDA remains at 100% across the area, with 2.5 meters near the window and a more elongated area of 1.6 meters in the narrower section where the skylights are located, achieving at least 80% sDA.

For sUDI, the parameters are similar to the single-path skylight but show slightly better lighting in the kitchen area.

Angled Skylight NE (Roof Monitor):

This scenario uses a single skylight in the same position as the Flat Skylight but placed vertically like a roof monitor. The sDA decreases to 92%, which still meets WELL standards, with some of the rear area falling below 50%.

For sUDI, the parameter drops slightly. The kitchen area receives daylight only within 1.4 meters.

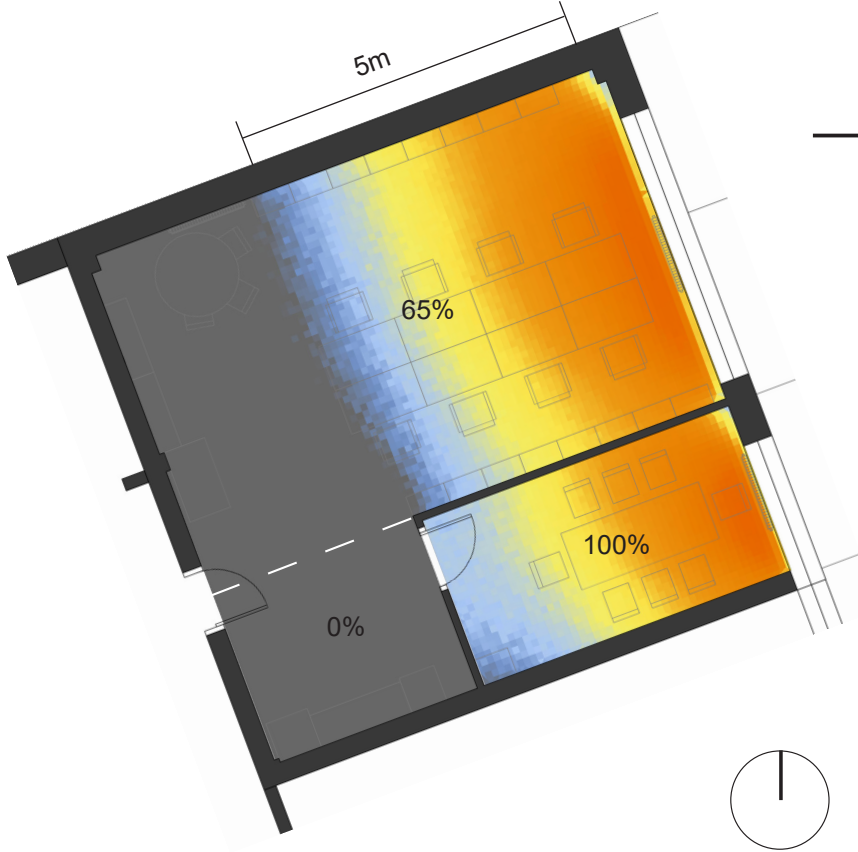
Angled Skylight N (Roof Monitor):

In this final scenario, the same skylight from the previous case is used, but it is oriented toward the north. This orientation produces similar results to the Angled Skylight NE but provides a better distribution of light, ensuring more even illumination across the space.

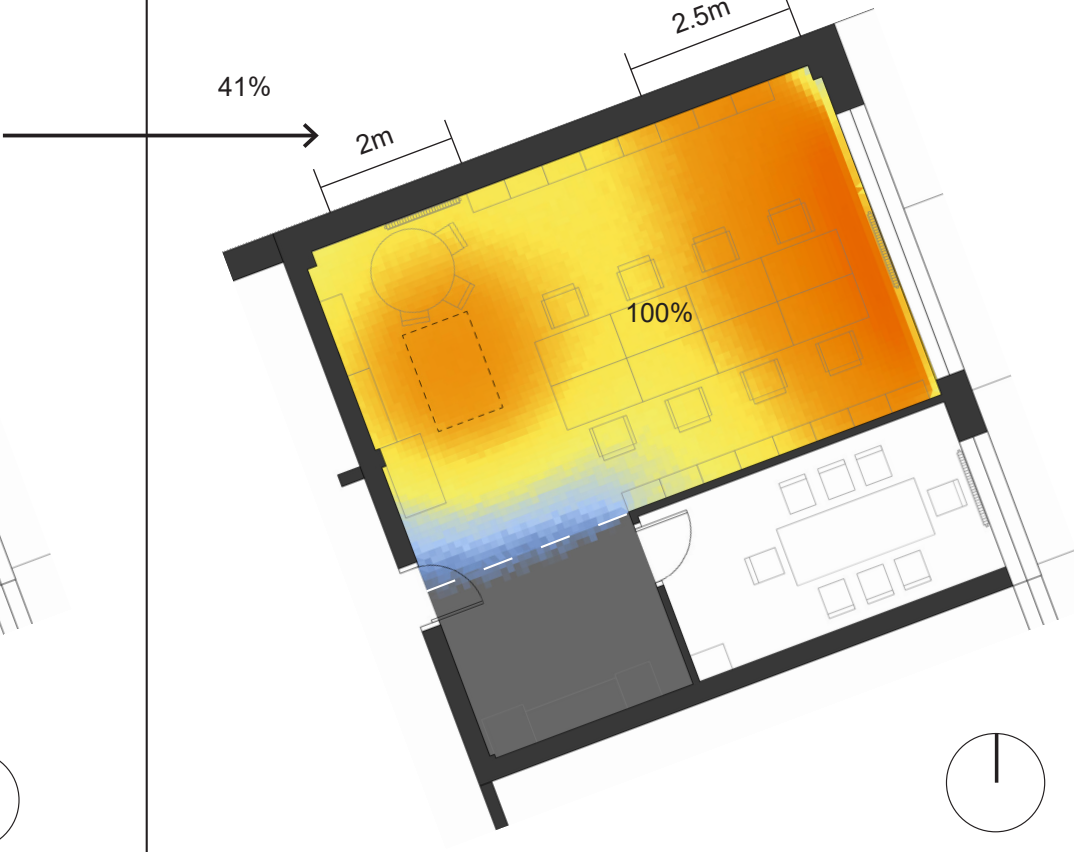
The flat skylights produce better results because the zenith of the sky is brighter than the horizon. However, the angled skylight reduces solar radiation gains, facilitates the opening of the skylight for ventilation, and provides sufficiently good daylighting performance.

Spatial Daylight Autonomy (sDA)
% area with ≥ 50% occupied hours ≥ 300 lux

CURRENT SITUATION

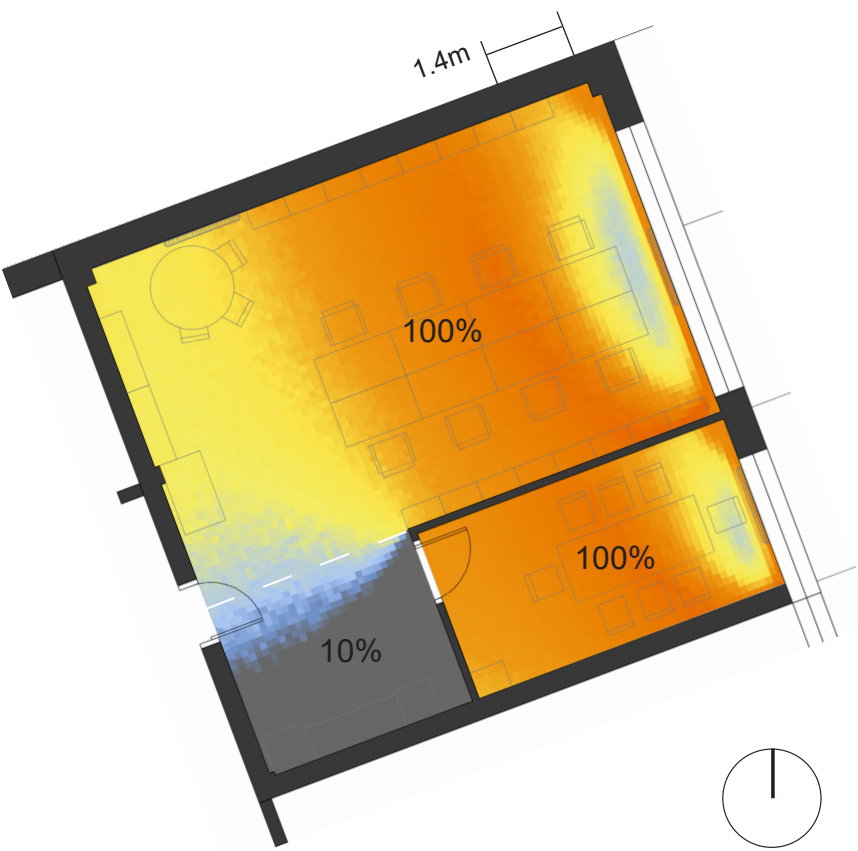


FLAT SKYLIGHT



Spatial Useful Daylight Illuminance (sUDI)
% area with ≥ 50% occupied hours 100>3000 lux

CURRENT SITUATION



FLAT SKYLIGHT

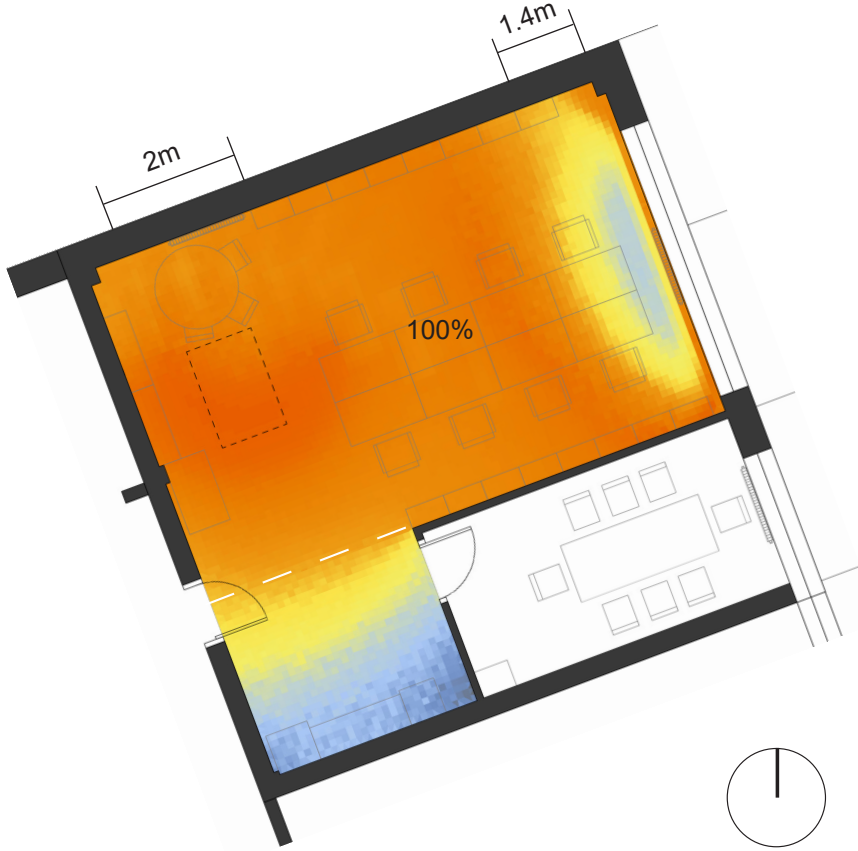
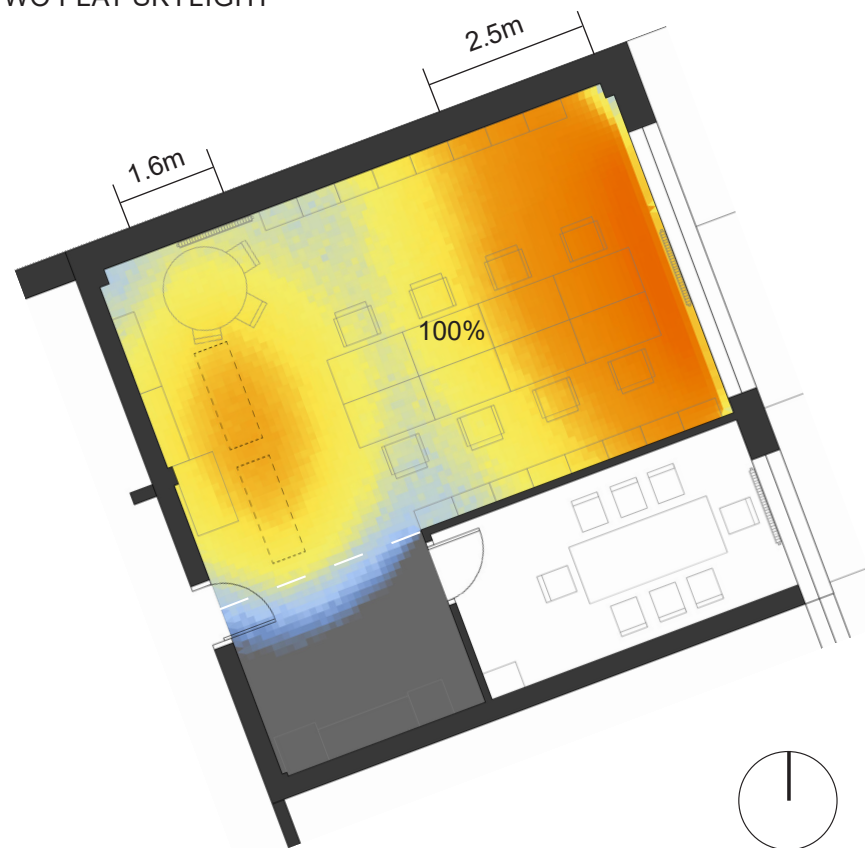


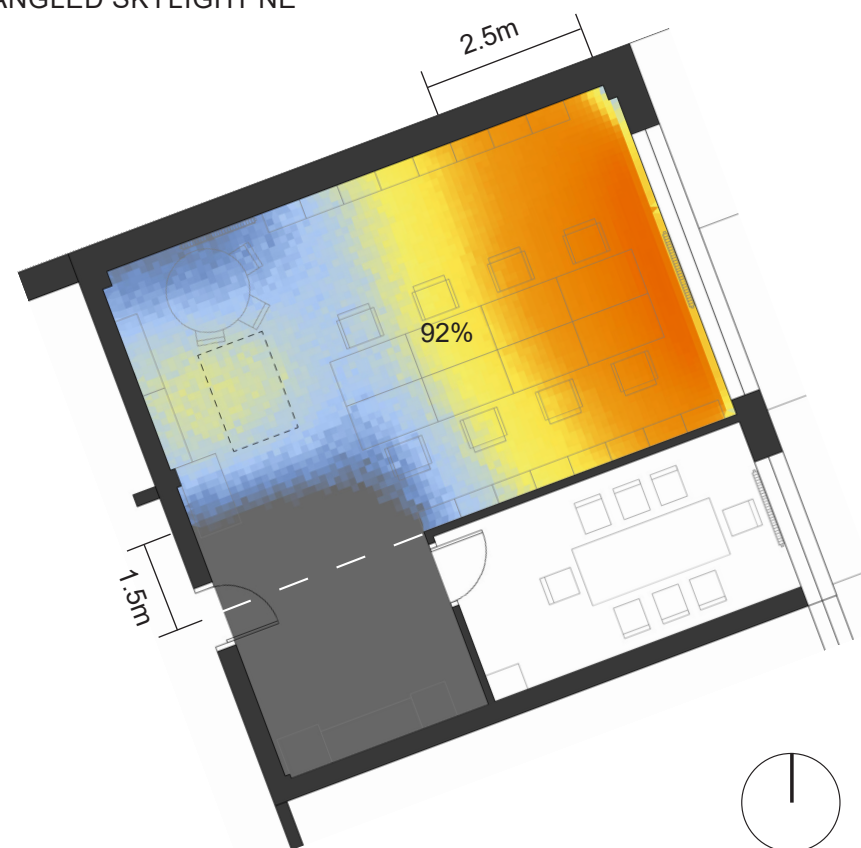
Fig. 75

Atmos Lab Office plan with the s DA and sUDI results for different skylight designs.

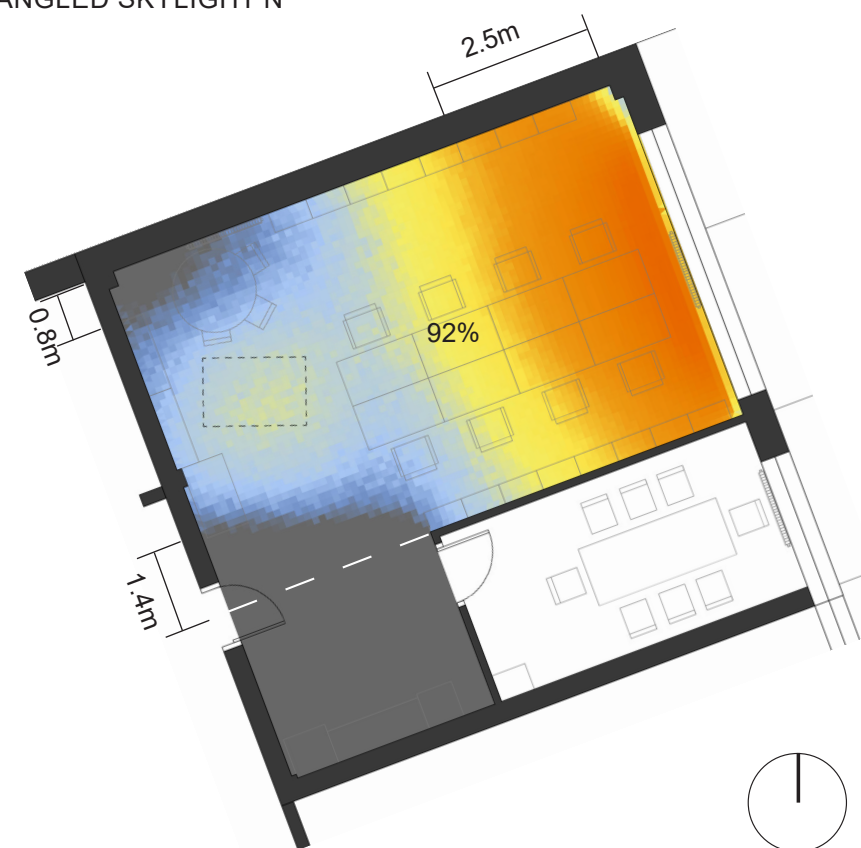
TWO FLAT SKYLIGHT



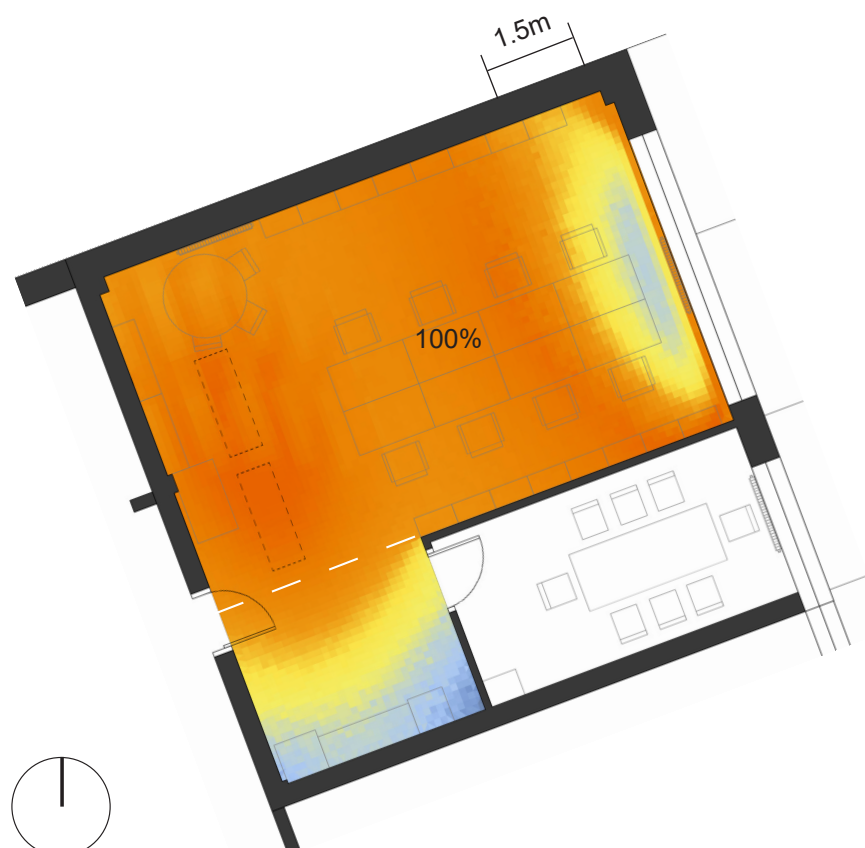
ANGLED SKYLIGHT NE



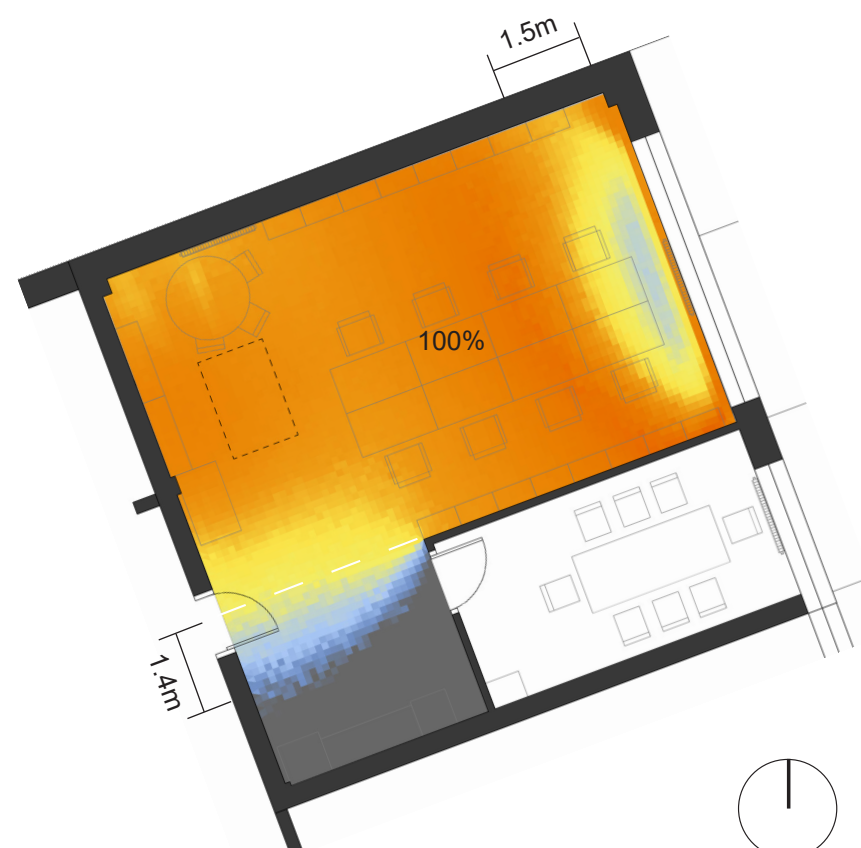
ANGLED SKYLIGHT N



TWO FLAT SKYLIGHT



ANGLED SKYLIGHT NE



ANGLED SKYLIGHT N

